

Geometric Interaction Graph Neural Network for Predicting Protein–Ligand Binding Affinities from 3D Structures (GIGN)

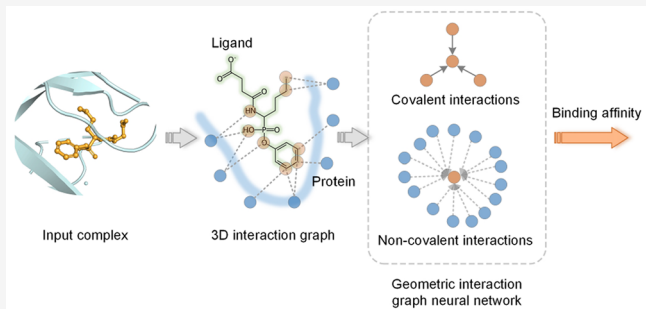
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ABSTRACT: Predicting protein–ligand binding affinities (PLAs) is a core problem in drug discovery. Recent advances have shown great potential in applying machine learning (ML) for PLA prediction. However, most of them omit the 3D structures of complexes and physical interactions between proteins and ligands, which are considered essential to understanding the binding mechanism. This paper proposes a geometric interaction graph neural network (GIGN) that incorporates 3D structures and physical interactions for predicting protein–ligand binding affinities. Specifically, we design a heterogeneous interaction layer that unifies covalent and noncovalent interactions into the message passing phase to learn node representations more effectively. The heterogeneous interaction layer also follows fundamental biological laws, including invariance to translations and rotations of the complexes, thus avoiding expensive data augmentation strategies. GIGN achieves state-of-the-art performance on three external test sets. Moreover, by visualizing learned representations of protein–ligand complexes, we show that the predictions of GIGN are biologically meaningful.



Drug discovery is an expensive process. Typically, a single drug can cost up to around 1 billion dollars and takes 10 years of development and testing before potentially being FDA approved.¹ Moreover, this process can fail at any point due to undesired side effects or experimental disproof of the promising therapeutic efficacy. A key problem in drug discovery is the prediction of the protein–ligand binding affinity (PLA), which describes the strength of the binding interaction between a ligand (drug candidate) and a protein (target). To identify effective and safe drugs for a given protein, pharmacologists have to test thousands of chemical compounds.² However, experimentally measuring the PLA is both time- and resource-consuming. In silico methods are able to accelerate the process of drug screening by ranking drug candidates and prioritizing the better ones according to the predictions of the PLA.

With the increasing amount of high-quality experimentally determined protein–ligand structures and their binding affinities, machine learning (ML) methods have been widely used to predict protein–ligand binding affinities or interactions by identifying useful patterns from data.^{2–7,7–27} Existing ML-based methods for PLA prediction can be divided into two categories, interaction-free and interaction-based methods, according to whether the models make decisions relying on physical interactions, as shown in Figure 1.

Interaction-free methods^{2,17–19,23–25,28–32} implicitly assume that ML models can predict PLA from data that do not reveal physical protein–ligand interactions. Therefore, ligands are often expressed by the simplified molecular-input line-entry

system (SMILES), or 2D graphs, and proteins are represented by sequences, while atomic interactions are omitted for simplicity. For example, DeepDTA¹⁷ uses SMILES strings and protein sequences as input from which two convolutional neural networks (CNNs) are used to extract features. On the other hand, GraphDTA¹⁸ and MGraphDTA² represent ligands as 2D molecule graphs to retain 2D structural information. The interaction-free methods can be used to predict binding affinities from complexes without 3D structure and physical interaction information. However, 3D structure and physical interaction priors have been demonstrated to be essential for improving the model generalization capability.³³

In contrast, interaction-based models^{3–7,7–16,34} make predictions based on the 3D structures of complexes and physical interactions of proteins and ligands. Among the interaction-based models, 3D-convolutional neural networks (3D-CNNs)^{5,12} and interaction graph neural networks (IGNNs)^{3,8,11,15,33} are most commonly used to predict binding affinities from 3D structures with atomic interaction information. 3D-CNNs express a complex as a 3D grid and extract the

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几何交互图神经网络用于从三维结构预测蛋白质-配体结合亲和力和力 (GIGN)

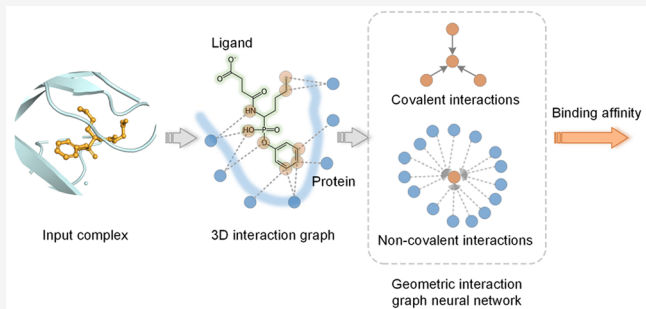
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摘要: 预测蛋白质–配体结合亲和力 (PLA) 是药物发现中的一个核心问题。最近的进展显示, 机器学习 (ML) 在 PLA 预测中的应用具有巨大潜力。然而, 绝大多数研究忽略了复合物的三维结构和蛋白质与配体之间的物理交互, 这被视为理解结合机制的必要因素。本文提出了一种几何交互图神经网络 (GIGN), 其结合了三维结构和物理交互来预测蛋白质–配体结合亲和力。具体而言, 我们设计了一个异质交互层, 将共价与非共价交互统一到信息传递阶段, 以更有效地学习节点表示。



异质交互层还遵循基本生物学家法则, 包括对复合物的平移和旋转的不变性, 从而避免了昂贵的数据增强策略。GIGN 在三个外部测试集上实现了最先进性能。此外, 通过可视化学习到的蛋白质–配体复合物的表示, 我们展示了 GIGN 的预测在生物学上具有意义。

药物发现^{rug} 是一个成本高昂的过程。

通常, 单一药物的研发费用可高达约10亿美元, 并且需要10年的开发和测试才能可能获得FDA批准。¹ 此外, 这一过程在任何阶段都可能因不良副作用或对有前景的治疗效果的实验性反驳而失败。药物发现中的一个关键问题是预测蛋白质–配体结合亲和力 (PLA), 该参数描述了配体 (药物候选物) 与蛋白质 (靶标) 之间结合相互作用的强度。为了识别对特定蛋白质有效且安全的药物, 药理学家必须测试数千种化合物。² 然而, 实验性地测量 PLA 既耗时又耗资源。计算机模拟方法能够通过药物候选物进行排名并优先筛选更优者, 从而加速药物筛选的过程。

随着高质量实验确定的蛋白质–小分子结构及其结合亲和力数量的增加, 机器学习 (ML) 方法已被广泛应用于预测蛋白质–小分子结合亲和力或通过识别数据中的有用模式来进行相互作用。^{2–7, 7–27} 现有的基于机器学习的 PLA 预测方法可以根据模型是否依赖于物理交互进行决策, 分为两类: 无交互和基于交互的方法, 如图 1 所示。

无交互方法^{2, 17–19, 23–25, 28–32} 隐含假设机器学习模型可以从不揭示物理蛋白质–配体相互作用的数据中预测 PLA。因此, 配体通常通过简化分子输入线条符号系统 (SMILES) 来表示,

而蛋白质则通过序列表示, 原子相互作用被为了简化而省略。例如, DeepDTA¹⁷ 使用 SMILES 字符串和蛋白质序列作为输入, 通过两个卷积神经网络 (CNNs) 提取特征。另一方面, GraphDTA¹⁸ 和 MGraphDTA² 将配体表示为二维分子图, 以保留二维结构信息。无交互方法可用于在没有三维结构和物理相互作用信息的情况下预测结合亲和力。但是, 已证明三维结构和物理相互作用先验对于提高模型的概括能力至关重要。³³

相反, 基于交互的模型^{3–7, 7–16, 34} 基于复合物的三维结构和蛋白质与配体的物理交互进行预测。在基于交互的模型中, 三维卷积神经网络 (3D-CNNs)^{5,12} 和交互图神经网络 (IGNNs)^{3,8,11,15,33} 最常用于从含有原子交互信息的三维结构中预测结合亲和力。3D-CNNs 将复合物表示为三维网格并提取

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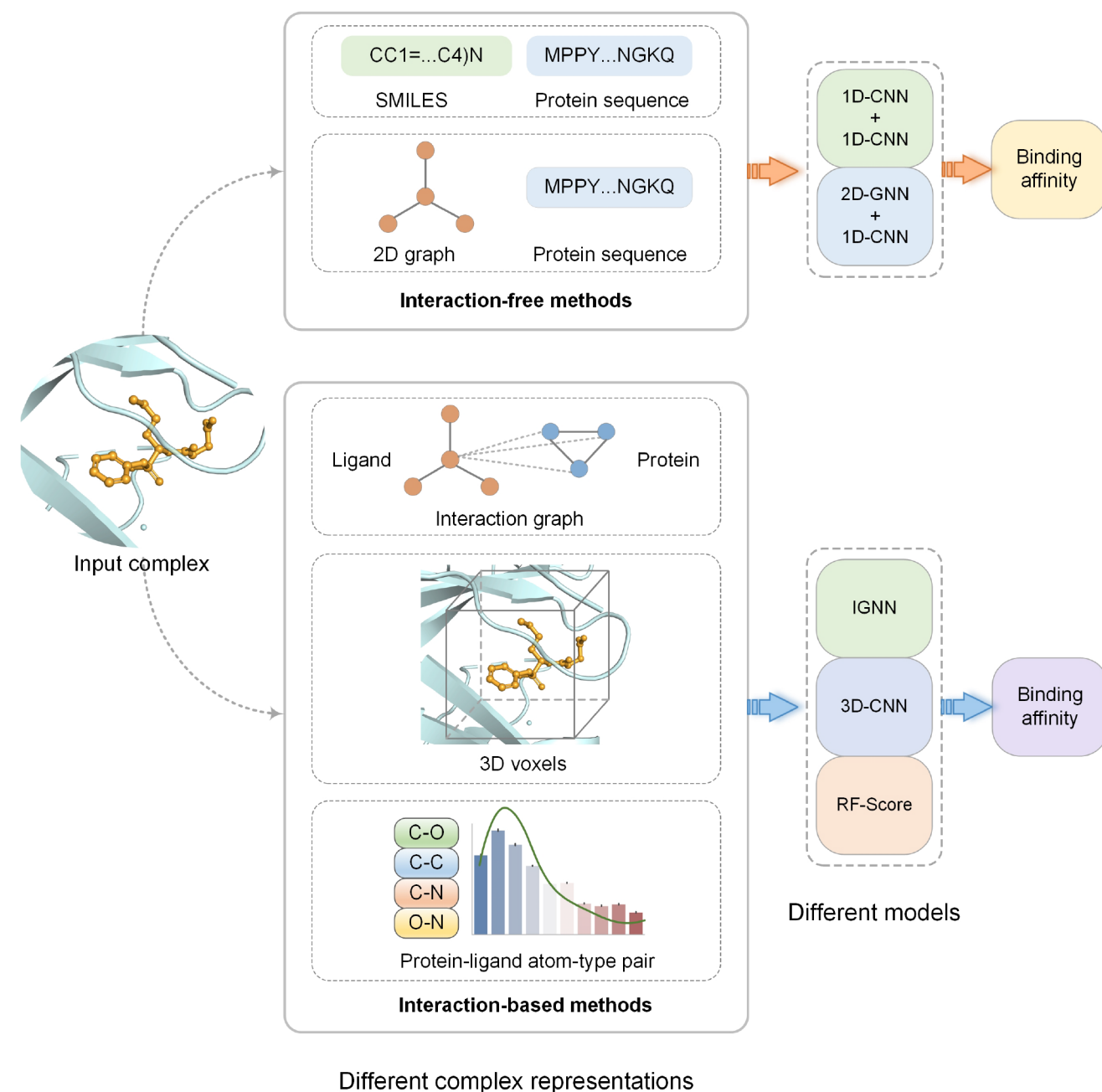


Figure 1. Interaction-free and interaction-based methods for PLA prediction.

structure and interaction information from the 3D grid using 3D convolutions, which is less efficient because most of the voxels do not contain useful information about the structure. Furthermore, since the position of each voxel is not rotationally invariant, rotating the atomic coordinates will change the prediction value of the binding affinity, which is inconsistent with biological fact. On the other hand, IGNNs represent protein–ligand complexes as interaction graphs, where nodes correspond to atoms and edges correspond to covalent or noncovalent bonds/interactions. The most common way to inject structure information into IGNNs is to use pairwise atom–atom distances computed from their 3D coordinates,^{3,11} which ensures IGNNs' invariance to translations and rotations of the complexes.

Despite the great potential of IGNNs in PLA prediction, they may not be ready for real applications due to their insufficient generalization ability.^{3,35} We summarized two potential problems of existing IGNNs in binding affinity prediction. First, existing IGNNs often regard covalent and noncovalent interactions as the same type of interaction.^{33,35–37} In this context, a ligand node can receive messages from its covalent and noncovalent neighbors simultaneously during message passing, as shown in Figure 2(a). Although this assumption simplifies the modeling, it has a significant disadvantage. The number of noncovalent interactions is greatly larger than that of covalent ones (see Supplement C for more details) and thus noncovalent interactions will dominate the computation (i.e., the information on covalent interactions may be engulfed by that on noncovalent ones). Second, the geometric prior (or

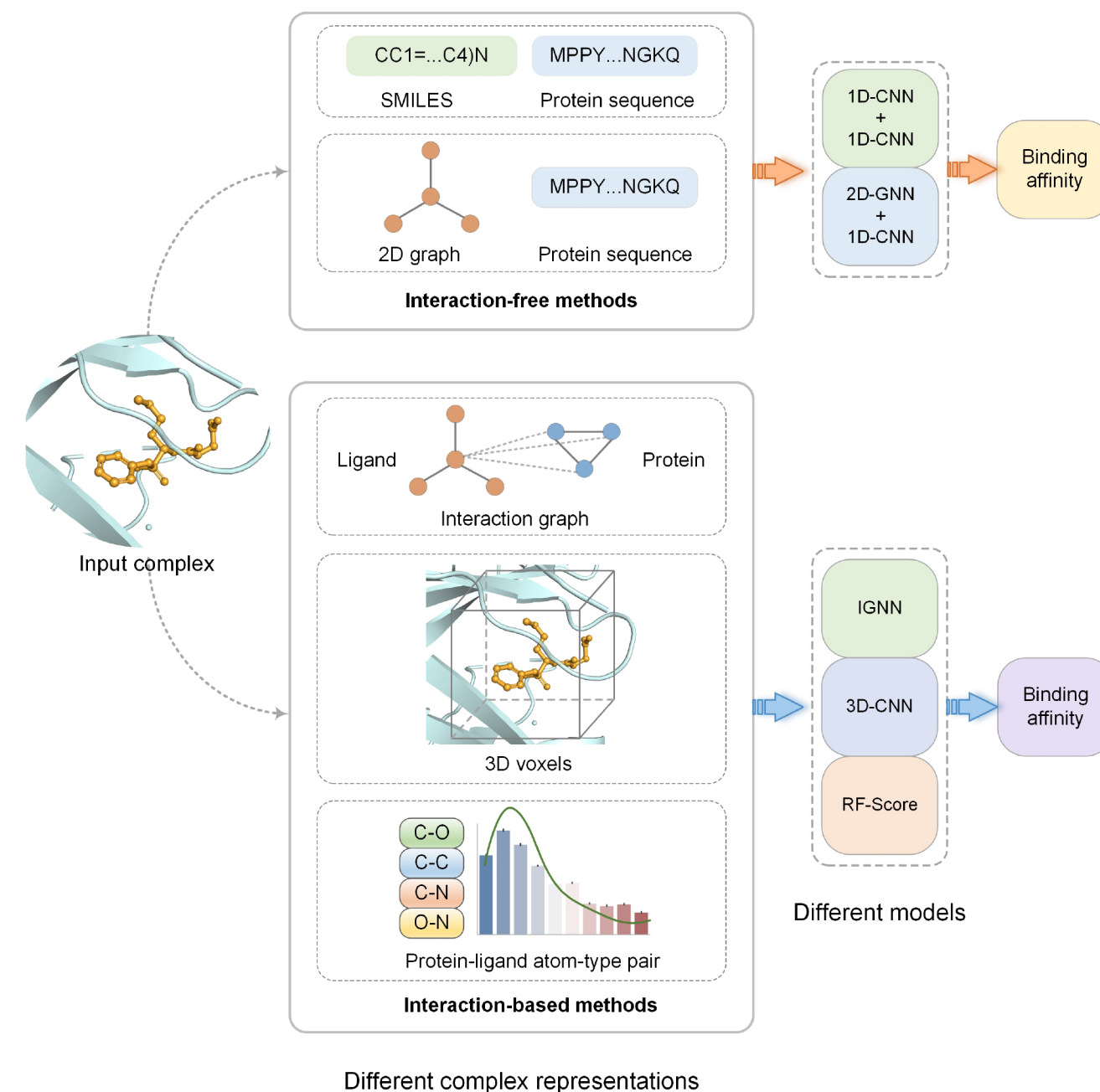


图 1. 无交互和基于交互的方法用于PLA 预测。

使用三维卷积从三维网格中提取结构和交互信息，这效率较低，因为大多数体素并不包含关于结构的有用信息。此外，由于每个体素的位置不是不变的，旋转原子坐标将改变结合亲和力的预测值，这与生物事实不一致。另一方面，IGNNs将蛋白质–配体复合物表示为交互图，其中节点对应于原子，边对应于共价或非共价键/交互。将结构信息注入IGNNs的最常见方法是使用从其三维坐标^{3,11}计算出的成对原子–原子距离，这确保了IGNNs对复合物的平移和旋转的不变性。

尽管 IGNNs 在 PLA 预测中具有巨大的潜力，但由于其不足的泛化能力，它们可能尚未准备好用于实际应用。^{3,35} 我们总结了现有 IGNNs 在结合亲和力预测中的两个潜在问题。首先，现有 IGNNs 通常将共价交互和非共价交互视为相同类型的交互。^{33,35–37} 在这种情况下，配体节点可以在消息传递过程中同时接收来自其共价和非共价邻居的消息，如图 2(a) 所示。尽管这一假设简化了建模，但它有一个显著的缺点。非共价交互的数量远大于共价交互的数量（见补充材料 C 以获取更多细节），因此非共价交互将在计算中占主导地位（即，关于共价交互的信息可能会被非共价交互的信息淹没）。其次，几何先验（或

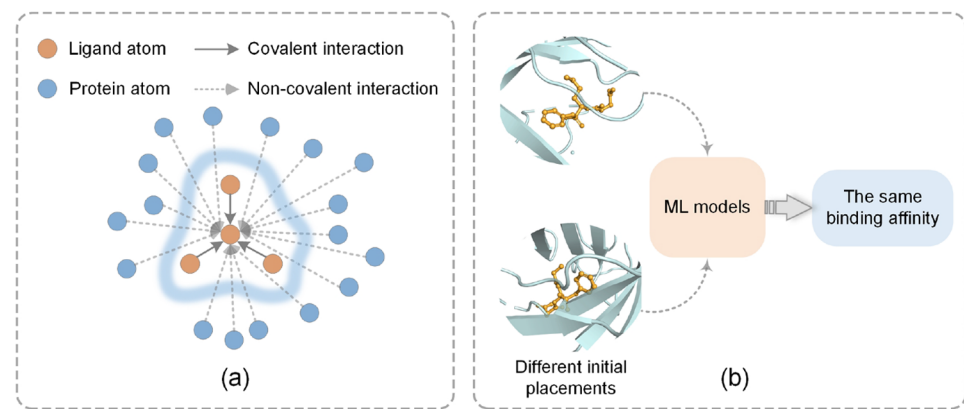


Figure 2. Motivation of the proposed method. (a) A ligand node receives messages from its covalent and noncovalent neighbors. In this example, the ligand node has 3 covalent neighbors and 17 noncovalent neighbors. Therefore, noncovalent interactions will dominate the computation. (b) The predicted binding affinity should be invariant to different initial 3D placements.

symmetry prior) is an important inductive bias that can be used to improve the model generalization ability by restricting neural networks to relevant functions through exploiting the symmetry of problems.^{1,38–40} For example, the predicted binding affinity should remain unchanged for the same protein–ligand complex with different initial placements (e.g., different positions and orientations), as shown in Figure 2(b). However, this invariance property still remains under-explored (e.g., there are fewer studies trying to demonstrate that IGNNs can benefit from considering invariance property).

To tackle the aforementioned issues, we propose a geometric interaction graph neural network (GIGN) that incorporates 3D structures and physical interactions with the invariance constraint to predict protein–ligand binding affinities. Key contributions of GIGN include the following:

- GIGN uses a heterogeneous interaction layer that unifies covalent and noncovalent interactions into the message passing phase to learn node representations more effectively. The heterogeneous interaction layer treats covalent and noncovalent interactions as different types of interactions and processes them independently during message passing, thus avoiding the drawback described in Figure 2(a).
- GIGN enforces the neural network to satisfy the invariance property with respect to input translations and rotations. We show that the model generalization ability can be greatly improved by considering the invariance property.
- The experimental results indicate that GIGN achieves state-of-the-art performance on three external test sets with a low computational cost, which is more readily applicable to large-scale databases.
- The visualization results show that GIGN can capture essential features relevant to binding.

PROBLEM DEFINITION

In this work, a complex is represented as a 3D interaction graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R}) = (\mathcal{V}_l \cup \mathcal{V}_p, \mathcal{E}_l \cup \mathcal{E}_p \cup \mathcal{E}_{lp}, \mathcal{R}_l \cup \mathcal{R}_p)$, where $\mathcal{V}_l$ and $\mathcal{V}_p$ are sets of ligand and protein nodes (i.e., atoms), respectively; $\mathcal{E}_l$ and $\mathcal{E}_p$ are sets of ligand and protein covalent edges (i.e., covalent bonds/interactions), respectively; $\mathcal{E}_{lp}$ is the set of noncovalent edges (i.e., noncovalent bonds/interactions); and $\mathcal{R}_l$ and $\mathcal{R}_p$ are sets of 3D coordinates of ligand

and protein nodes, respectively. Each node $v_i \in \mathcal{V}$ has its initial feature $x_i \in \mathbb{R}^n$ and 3D coordinate $r_i \in \mathbb{R}^3$. A covalent edge $e_{ij} \in \mathcal{E}_l \cup \mathcal{E}_p$ exists when atoms v_i and v_j are connected by covalent bonds. A noncovalent edge $e_{ik} \in \mathcal{E}_{lp}$ exists when $d_{ik} < 5 \text{ \AA}$, where $d_{ik} = \|r_i - r_k\|^2$ is the distance between the ligand atom $v_i \in \mathcal{V}_l$ and the protein atom $v_k \in \mathcal{V}_p$. The initial features used for nodes can be found in Supplement A Table S1. Given a 3D interaction graph, our goal is to learn a model f that satisfies invariance to predict the binding affinity as $f(\mathcal{G}) = \hat{y}$, where $\hat{y} \in \mathbb{R}^1$.

GRAPH NEURAL NETWORKS

Graph neural networks (GNNs) are permutation-equivariant networks that operate on graph data.^{39,41} Specifically, a GNN model uses a message passing function⁴¹ to iteratively update the representation of a node by aggregating messages from its neighboring nodes according to

$$m_{ji} = M(h_j^{(t)}, h_i^{(t)}) \quad (1)$$

$$m_i = \sum_{v_j \in \mathcal{N}(v_i)} m_{ji} \quad (2)$$

$$h_i^{(t+1)} = U(h_i^{(t)}, m_i) \quad (3)$$

where $h_i^{(t)}$ denotes the node representation of v_i at t -th layer of GNNs, $h_i^{(0)} = Wx_i$ ($W \in \mathbb{R}^{m \times n}$), $\mathcal{N}(v_i)$ is the set of neighbors of v_i , and M and U are differentiable functions such as multilayer perceptrons (MLPs).

INVARIANCE

Let $T: \mathcal{X} \rightarrow \mathcal{X}$ be a set of transformations (e.g., rotation, translation, reflection, or permutation). A graph neural network $f: \mathcal{X} \rightarrow \mathcal{Y}$ is invariant to T on $\mathcal{X}$ if

$$f(T(X)) = f(X) \quad (4)$$

In this work we explore the following two types of invariance:

- Translation invariance, where translating the input by a translation vector g results in an invariant output (i.e., $f(X + g) = f(X)$).
- Rotation invariance, where rotating the input by a rotation matrix Q results in an invariant output (i.e., $f(QX) = f(X)$).

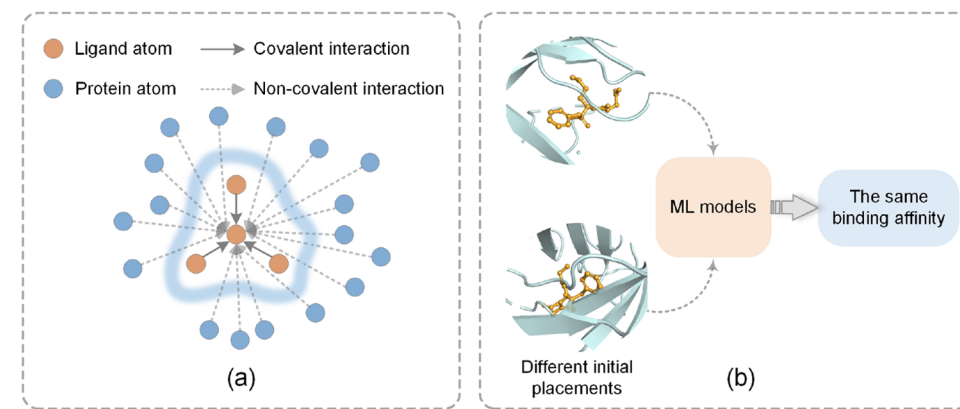


图 2. 提出方法的动机。(a) 配体节点接收来自其共价和非共价邻居的消息。在此示例中，配体节点有3个共价邻居和17个非共价邻居。因此，非共价交互将在计算中占主导地位。(b) 预测结合亲和力应对不同的初始三维放置保持不变。

对称先验是一个重要的归纳偏差，可以通过利用问题的对称性限制神经网络到相关功能，从而提高模型泛化能力。^{138–40} 例如，对于相同的蛋白质–配体复合物，不同的初始位置（例如，不同的位置和方向）下，预测结合亲和力应保持不变，如图 2(b)所示。然而，这种不变性特性仍然未得到充分探索（例如，针对 IGNNs 通过考虑不变性特性获益的研究较少）。

为了解决上述问题，我们提出了一种几何交互图神经网络（GIGN），该网络结合了三维结构和物理交互，并引入不变性约束以预测蛋白质–配体结合亲和力。GIGN的主要贡献包括以下几点：

- GIGN采用了一种异质交互层，将共价和非共价交互统一纳入信息传递阶段，以更有效地学习节点表示。异质交互层将共价和非共价交互视为不同类型的交互，并在信息传递过程中独立处理它们，从而避免了在图 2(a)中描述的缺陷。
- GIGN强制神经网络满足关于输入平移和旋转的不变性特性。我们展示了，通过考虑不变性特性，模型的泛化能力可以得到显著提高。
- 实验结果表明，GIGN 在三个外部测试集上实现了最先进性能，且具有低计算成本，这使其更易于应用于大规模数据库。
- 可视化结果显示，GIGN 能够捕捉与结合相关的基本特征。

问题定义

在本研究中，一个复合物被表示为一个三维交互图，其中和分别是配体和蛋白质节点（即原子）的集合；和分别是配体和蛋白质共价边（即共价键/相互作用）的集合；是非共价边（即非共价键/相互作用）的集合；和分别是配体和蛋白质节点的三维坐标集合。

$\mathcal{V}_l$

1

p

每个节点具有其初始特征和三维坐标。当原子 v_i 和 v_j 通过共价键连接时，存在一个共价边 e_{ij} 。当 $d_{ik} < 5 \text{ \AA}$ 时，存在一个非共价边 e_{ik} 。

e_{ik} $\mathcal{E}_{lp}$

其中 $d_{ik} = \|r_i - r_k\|^2$ 是配体原子与蛋白质原子之间的距离。用于节点的初始特征可以在补充材料 A 表 S1 中找到。给定一个三维交互图，我们的目标是学习一个模型 f ，该模型满足不变性，以预测结合亲和力，如，其中 $\hat{y} = f(\mathcal{G})$ 。

图神经网络

图神经网络（GNNs）是对图数据进行操作的置换等价网络。^{39,41} 具体而言，GNN模型使用消息传递函数⁴¹通过聚合来自其邻近节点的消息，迭代更新节点的表示，遵循

$$m_{ji} = M(h_j^{(t)}, h_i^{(t)}) \quad (1)$$

$$m_i = \sum_{v_j \in \mathcal{N}(v_i)} m_{ji} \quad (2)$$

$$h_i^{(t+1)} = U(h_i^{(t)}, m_i) \quad (3)$$

其中 $h_i^{(t)}$ 表示 v_i 在 GNNs 的 t -层的节点表示， $h_i^{(0)} = Wx_i$ 是 v_i 的邻居集合， M 和 U 是可微分函数，如多层感知器（MLPs）。

不变性

设有 T -组变换（例如，旋转、平移、反射或置换）。若图神经网络对 T 在 $\mathcal{X}$ 上不变， f ：

$$f(T(X)) = f(X) \quad (4)$$

在本研究中，我们探讨以下两种类型的不变性：

- 平移不变性，其中通过平移向量 g 对输入进行平移的结果是一个不变的输出（即， $f(X+g) = f(X)$ ）。
- 旋转不变性，其中通过旋转矩阵 Q 对输入进行旋转会产生不变的输出（即， $f(QX) = f(X)$ ）。

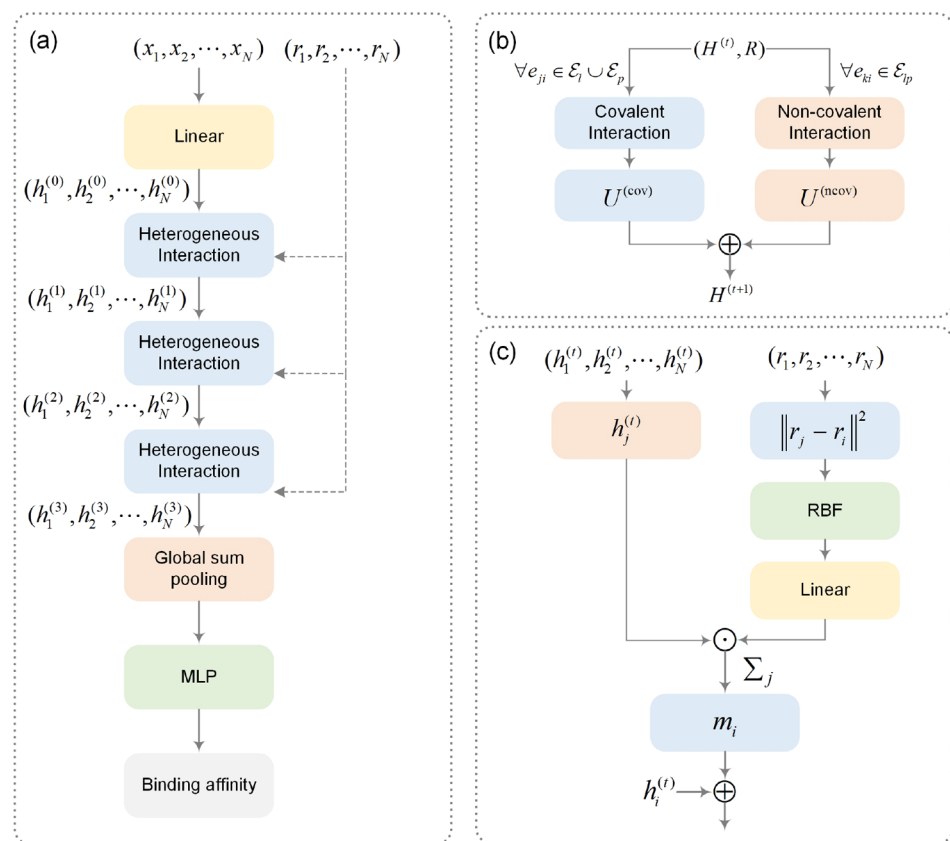


Figure 3. (a) The computational pipeline of GIGN. (b) The heterogeneous interaction layer. (c) The details of covalent and noncovalent interactions of the heterogeneous interaction layer.

GEOMETRIC INTERACTION GRAPH NEURAL NETWORK

The overall computational pipeline of GIGN is depicted in Figure 3(a). GIGN uses a heterogeneous interaction layer that unifies covalent and noncovalent interactions into the message passing phase to learn node representations more effectively, as shown in Figure 3(b). Specifically, for a 3D interaction graph $\mathcal{G}$ with node features $(x_1, x_2, \dots, x_N)$ and 3D coordinates $(r_1, r_2, \dots, r_N)$, the heterogeneous interaction layer can be defined as

$$m_{ji}^{(\text{cov})} = M^{(\text{cov})}(h_j^{(t)}, h_i^{(t)}, \lambda(\|r_j - r_i\|^2)), \forall e_{ji} \in \mathcal{E}_1 \cup \mathcal{E}_p \quad (5)$$

$$m_{ki}^{(\text{ncov})} = M^{(\text{ncov})}(h_k^{(t)}, h_i^{(t)}, \lambda(\|r_k - r_i\|^2)), \forall e_{ki} \in \mathcal{E}_{\text{lp}} \quad (6)$$

$$m_i^{(\text{cov})} = \sum_{v_j \in \mathcal{N}(v_i)} m_{ji}^{(\text{cov})}, \forall e_{ji} \in \mathcal{E}_1 \cup \mathcal{E}_p \quad (7)$$

$$m_i^{(\text{ncov})} = \sum_{v_k \in \mathcal{N}(v_i)} m_{ki}^{(\text{ncov})}, \forall e_{ki} \in \mathcal{E}_{\text{lp}} \quad (8)$$

$$h_i^{(t+1)} = U^{(\text{cov})}(h_i^{(t)}, m_i^{(\text{cov})}) + U^{(\text{ncov})}(h_i^{(t)}, m_i^{(\text{ncov})}) \quad (9)$$

where $M^{(\text{cov})}$ and $M^{(\text{ncov})}$ are implemented as $M^{(\text{cov})}(h_j^{(t)}, h_i^{(t)}, \lambda(\|r_j - r_i\|^2)) = h_j^{(t)} \odot [\lambda(\|r_j - r_i\|^2)]$ and $M^{(\text{ncov})}(h_k^{(t)}, h_i^{(t)}, \lambda(\|r_k - r_i\|^2)) = h_k^{(t)} \odot [\lambda(\|r_k - r_i\|^2)]$ ($\odot$ is the element-wise product), respectively, as shown in Figure 3(c), $U^{(\text{cov})}$ and $U^{(\text{ncov})}$ are implemented as MLPs, and λ is the

linear combination of Gaussian radial basis functions (RBFs)⁴² $\exp(-\frac{1}{\sigma^2} \|d_{ji} - \mu_k\|^2)$ located at centers $0 \text{ \AA} \leq \mu_k \leq 6 \text{ \AA}$ every $2/3 \text{ \AA}$ with $\sigma = 2/3 \text{ \AA}$. This is chosen to guarantee all distances within $5 \text{ \AA}$ are covered. Through eqs 5 and (6), we inject the structure information $\lambda(\|r_j - r_i\|^2)$ and $\lambda(\|r_k - r_i\|^2)$ into the model. Intuitively, $m_{ji}^{(\text{cov})}$ and $m_{ki}^{(\text{ncov})}$ are at different scales since the number of noncovalent interactions is greatly larger than that of covalent interactions. Therefore, we use $U^{(\text{cov})}$ and $U^{(\text{ncov})}$ to convert $m_{ji}^{(\text{cov})}$ and $m_{ki}^{(\text{ncov})}$ into the same scale before integrating them, as shown in eq 9, which avoids the issue described by Figure 2 (a). Note that the noncovalent interactions within the ligand (or the protein) are also included because the receptive field of GNNs can be expanded with increasing layers^{2,43} (i.e., an atom can access its k -hop neighbors for a k -layer GNN).

ANALYSIS ON INVARIANCE FOR GIGN

Note that the heterogeneous interaction layer F is invariant to translations and rotations. Formally, F satisfies

$$H^{(t+1)} = F(R, H^{(t)}) = F(QR + g, H^{(t)}) \quad (10)$$

where $H = (h_1, \dots, h_N) \in \mathbb{R}^{N \times m}$ is the node representation matrix of a graph with N nodes, $R = (r_1, \dots, r_N) \in \mathbb{R}^{N \times 3}$ is the 3D coordinate matrix, $g \in \mathbb{R}^3$ is a translation vector, and $Q \in \mathbb{R}^{3 \times 3}$ is a rotation matrix (orthogonal matrix).

The initial node feature matrix $X = (x_1, \dots, x_N) \in \mathbb{R}^{N \times n}$ is invariant to transformations (translations and rotations) on R ; in other words, we do not inject any position information about R

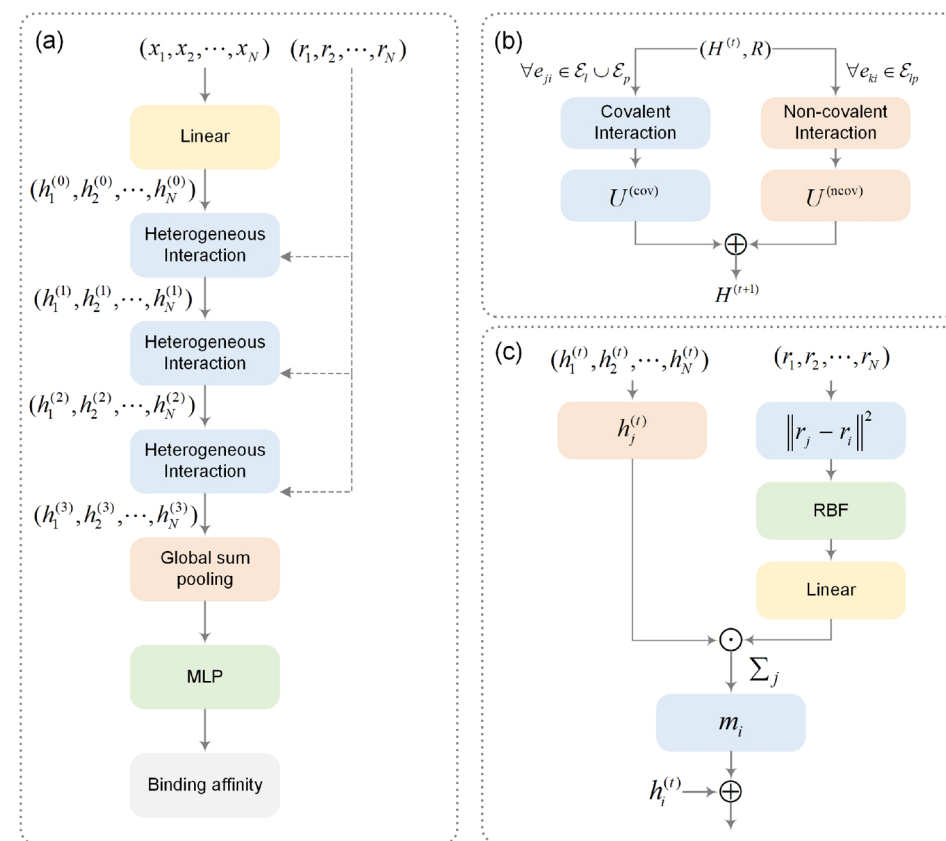


图 3. (a) GIGN的计算流程。 (b) 异质交互层。 (c) 异质交互层中共价和非共价交互的细节。

几何交互图神经网络

GIGN的整体计算流程如图 3(a)所示。GIGN使用一个异质交互层，将共价交互和非共价交互统一到信息传递阶段，以更有效地学习节点表示，如图 3(b)所示。具体而言，对于一个具有节点特征 $(x_1, x_2, \dots, x_N)$ 和三维坐标 $(r_1, r_2, \dots, r_N)$ 的三维交互图，异质交互层可以定义为

$$m_{ji}^{(\text{cov})} = M^{(\text{cov})}(h_j^{(t)}, h_i^{(t)}, \lambda(\|r_j - r_i\|^2)), \forall e_{ji} \in \mathcal{E}_1 \cup \mathcal{E}_p \quad (5)$$

$$m_{ki}^{(\text{ncov})} = M^{(\text{ncov})}(h_k^{(t)}, h_i^{(t)}, \lambda(\|r_k - r_i\|^2)), \forall e_{ki} \in \mathcal{E}_{\text{lp}} \quad (6)$$

$$m_i^{(\text{cov})} = \sum_{v_j \in \mathcal{N}(v_i)} m_{ji}^{(\text{cov})}, \forall e_{ji} \in \mathcal{E}_1 \cup \mathcal{E}_p \quad (7)$$

$$m_i^{(\text{ncov})} = \sum_{v_k \in \mathcal{N}(v_i)} m_{ki}^{(\text{ncov})}, \forall e_{ki} \in \mathcal{E}_{\text{lp}} \quad (8)$$

$$h_i^{(t+1)} = U^{(\text{cov})}(h_i^{(t)}, m_i^{(\text{cov})}) + U^{(\text{ncov})}(h_i^{(t)}, m_i^{(\text{ncov})}) \quad (9)$$

在 $M^{(\text{cov})}$ 和 $M^{(\text{ncov})}$ 分别实现为 $\odot$ 元素级乘积，如图 3(c)所示， $U^{(\text{cov})}$ 和 $U^{(\text{ncov})}$ 被实现为多层感知器，且 λ 是 $\odot$ 和 $\odot$ 的线性组合。GIGN 的不变性分析：注意，异质交互层 F 对于平移和旋转是不变的。从形式上来看， F 满足

高斯径向基函数 (RBFs)⁴² 的线性组合位于中心 $0 \text{ \AA} \leq \mu_k \leq 6 \text{ \AA}$ 每 $2/3 \text{ \AA}$ 与 $\sigma = 2/3 \text{ \AA}$ 。这是为了确保在 $5 \text{ \AA}$ 内的所有距离都被覆盖。通过公式 5 和 (6)，我们将结构信息 $\lambda(\|r_j - r_i\|^2)$ 和 $\lambda(\|r_k - r_i\|^2)$ 注入模型中。从直观上看， $m_{ji}^{(\text{cov})}$ 和 $m_{ki}^{(\text{ncov})}$ 处于不同的尺度，因为非共价交互的数量远大于共价交互的数量。因此，我们使用 $U^{(\text{cov})}$ 和 $U^{(\text{ncov})}$ 将 $m_{ji}^{(\text{cov})}$ 和 $m_{ki}^{(\text{ncov})}$ 转换为相同的尺度，然后进行整合，如公式 9 所示，这避免了图 2 (a) 中描述的问题。请注意，配体（或蛋白质）内的非共价交互也被包含在内，因为图神经网络的感受野可以通过增加层数而扩展^{2,43}（即，原子可以访问其 k -hop 邻居，对于 k 层图神经网络）。

对GIGN的不变性分析

请注意，异质交互层 F 对于平移和旋转是不变的。从形式上来看， F 满足

$$H^{(t+1)} = F(R, H^{(t)}) = F(QR + g, H^{(t)}) \quad (10)$$

其中是具有 N 个节点的图的节点表示矩阵，是三维坐标矩阵，是平移向量，以及是旋转矩阵（正交矩阵）。

初始节点特征矩阵对于在 R 上的变换（平移和旋转）是不可变的；换句话说，我们并未注入关于 R 的任何位置信息。

into X . Therefore, $H^{(0)} = WX$ ($W \in \mathbb{R}^{m \times n}$) is also invariant to transformations. Given that distance between two nodes is invariant to translations $\|(r_j + g) - (r_i + g)\|^2 = \|r_j - r_i\|^2$ and to rotations $\|Qr_j - Qr_i\|^2 = [Q(r_j - r_i)]^T [Q(r_j - r_i)]$, we can

$$\begin{aligned} &= (r_j - r_i)^T Q^T Q (r_j - r_i) \\ &= \|r_j - r_i\|^2 \end{aligned}$$

conclude that the message $m_{ji}^{(\text{cov})}$ in eq 5 is also invariant:

$$\begin{aligned} m_{ji}^{(\text{cov})} &= M^{(\text{cov})}(h_j^{(t)}, h_i^{(t)}, \lambda(\|(Qr_j + g) - (Qr_i + g)\|^2)) \\ &= M^{(\text{cov})}(h_j^{(t)}, h_i^{(t)}, \lambda(\|r_j - r_i\|^2)) \end{aligned} \quad (11)$$

Similarly, $m_{ji}^{(\text{ncov})}$ in eq 6 is invariant. Therefore, $m_i^{(\text{cov})}$, $m_i^{(\text{ncov})}$, and $h_i^{(t+1)}$ in eqs 7–9 are also invariant because they only depend on $m_{ji}^{(\text{cov})}$, $m_{ji}^{(\text{ncov})}$, and $h_i^{(t)}$, respectively. Thus, we can conclude that $F(QR + g, H^{(t)}) = F(R, H^{(t)})$.

BINDING AFFINITY PREDICTION

We use a global sum pool to encapsulate the node-level representations at the last graph convolution layer (T -th layer) into a graph-level vector as

$$h_G = \sum_{v_i \in V} h_i^{(T)} \quad (12)$$

An MLP ($\mathbb{R}^m \rightarrow \mathbb{R}^1$) is used to map h_G to a scalar $\hat{y} \in \mathbb{R}^1$ indicating the binding affinity. The model is trained to minimize a mean-squared error loss function as follows:

$$L_{\text{mse}} = \frac{1}{|\mathcal{D}|} \sum_{n=1}^{|\mathcal{D}|} (\hat{y}_n - y_n)^2 \quad (13)$$

where y_n is the experimental binding affinity of n -th sample in data set $\mathcal{D}$ and $|\mathcal{D}|$ is the sample size.

IMPLEMENTATION DETAIL

Experiments are conducted using an NVIDIA GeForce RTX A4000 with 16 GB memory. An Adam optimizer with a 5×10^{-4} learning rate is used to update model parameters. We apply a weight decay of 1×10^{-6} for GIGN to reduce overfitting. The batch size is set to 128. The source code for GIGN is available at <https://github.com/guaguabujianle/GIGN>.

DATA SET PREPARATION

The proposed GIGN is trained and validated on PDBbind ver. 2016. 13 285 complexes (general and refined sets) with experimental binding affinities expressed as pK_a ($-\log K_d$ or $-\log K_i$) values are collected from the PDBbind website (<http://www.pdbbind.org.cn/>). We remove 381 samples that can not be processed by RDkit. The remaining 12904 samples are randomly split into training ($N = 11904$) and validation ($N = 1000$) sets. Three independent external test sets, the PDBbind 2013 core set ($N = 107$), the 2016 core set ($N = 285$), and the 2019 holdout set ($N = 4366$), are used to test the generalization capability of GIGN. The 2013 and 2016 core sets are two commonly used benchmarks to evaluate the performance of binding affinity prediction.^{3,5,10,14,35} However, their small sample sizes tend to result in overly optimistic results.⁴ Therefore, we collect 4366 samples from PDBbind ver. 2019

that are unavailable in the other four sets as a new external holdout set, mimicking a real temporal split scenario in which binding affinities for newly released structures are predicted by a model trained on past structural data. This temporal split setting is a more difficult task and also more precisely reflects the real situation of drug discovery.⁴⁴ Note that there is no overlap of the complexes between the five sets. To demonstrate that GIGN can be generalized to the non-PDBbind data set, we also validate it in the CSAR-HiQ data set.⁴⁵ The CSAR-HiQ data set contains two subsets with 343 complexes in total. After removing the complexes that overlap with the above five sets and can not be processed by RDkit, we retain 47 samples as a non-PDBbind external test set. During the processing, ligand–protein atom pairs with a distance greater than 5 Å are removed, following previous studies.^{11,33}

PERFORMANCE INDICATORS

The model performance is evaluated by Pearson's correlation coefficient (R_p) and the root-mean-square error (RMSE), which are defined as follows.

$$\text{RMSE} = \sqrt{\frac{1}{|\mathcal{D}|} \sum_{n=1}^{|\mathcal{D}|} (\hat{y}_n - y_n)^2} \quad (14)$$

$$R_p = \frac{\sum_{i=1}^{|\mathcal{D}|} (\hat{y}_n - \bar{\hat{y}})(y_n - \bar{y})}{\sqrt{\sum_{i=1}^{|\mathcal{D}|} (\hat{y}_n - \bar{\hat{y}})^2 (y_n - \bar{y})^2}} \quad (15)$$

BASELINES

We compare GIGN with the following representative methods.

- Interaction-free methods:
 - DeepDTA¹⁷ represents ligands and proteins as SMILES and sequences, respectively, and establishes two CNNs to extract features from SMILES and sequences, respectively.
 - GraphDTA¹⁸ is similar to DeepDTA but represents ligands as 2D graphs and uses GNNs (including GCN, GAT, GIN, and GAT-GCN) to learn representations from graphs.
 - MGraphDTA² also represents ligands as 2D graphs and uses a multiscale GNN to learn representations from graphs.
- Interaction-based methods:
 - Pafnucy⁵ is a representative 3D-CNN model that can learn the spatial structure and physical interactions of protein–ligand complexes.
 - OnionNet¹⁴ generates interaction-based features based on contact numbers in protein–ligand interaction shells and uses 2D-CNN to learn representations for prediction.
 - PotentialNet¹⁵ presents a staged gated graph neural network to include both intramolecular interactions and noncovalent interactions between ligands and proteins.
 - GNN-DTI¹¹ is a GNN-based method that incorporates the 3D structural information and uses distance-aware graph attention to differentiate the contribution of each interaction to the binding affinity.

物理化学快报

进入 X 。因此, $H^{(0)} = WX$ 对于变换也是不变的。考虑到两个节点之间的距离对平移是不变的 $\|(r_j + g) - (r_i + g)\|^2 = \|r_j - r_i\|^2$ 和

$$\begin{aligned} \text{rotations } \|Qr_j - Qr_i\|^2 &= [Q(r_j - r_i)]^T [Q(r_j - r_i)], \text{ we can} \\ &= (r_j - r_i)^T Q^T Q (r_j - r_i) \\ &= \|r_j - r_i\|^2 \end{aligned}$$

得出该信息 $m_{ji}^{(\text{cov})}$ 在公式 5中也是不变的:

$$\begin{aligned} m_{ji}^{(\text{cov})} &= M^{(\text{cov})}(h_j^{(t)}, h_i^{(t)}, \lambda(\|(Qr_j + g) - (Qr_i + g)\|^2)) \\ &= M^{(\text{cov})}(h_j^{(t)}, h_i^{(t)}, \lambda(\|r_j - r_i\|^2)) \end{aligned} \quad (11)$$

类似地, $m_{ji}^{(\text{ncov})}$ 在公式 6 中是不变的。因此, $m_i^{(\text{cov})}$ 、 $m_i^{(\text{ncov})}$ 和 $h_i^{(t+1)}$ 在公式 7–9 中也是不变的, 因为它们仅依赖于 $m_{ji}^{(\text{cov})}$ 、 $m_{ji}^{(\text{ncov})}$ 和 $h_i^{(t)}$, 因此我们可以得出结论 $F(QR + g, H^{(t)}) = F(R, H^{(t)})$ 。

结合亲和力预测

我们使用全局求和池将最后一层图卷积层 (T -th 层) 的节点级表示封装成图级向量, 如

$$h_G = \sum_{v_i \in V} h_i^{(T)} \quad (12)$$

使用 MLP (多层感知器) 将其映射到一个标量, 以指示结合亲和力。该模型经过训练, 最小化以下均方误差损失函数:

$$L_{\text{mse}} = \frac{1}{|\mathcal{D}|} \sum_{n=1}^{|\mathcal{D}|} (\hat{y}_n - y_n)^2 \quad (13)$$

其中 y_n 是数据集中第 n 个样本的实验结合亲和力, 且为样本大小。

实施细节

实验是在具有 16 GB 内存的 NVIDIA GeForce RTX A4000 上进行的。使用带有 5×10^{-4} 学习率的 Adam 优化器来更新模型参数。我们应用一个 1×10^{-6} 的权重衰减来减少 GIGN 的过拟合。批量大小设定为 128。GIGN 的源代码可在<https://github.com/guaguabujianle/GIGN>找到。

数据集准备

所提出的 GIGN 在 PDBbind 2016版 上进行了训练和验证。共有 13 285 个复合物 (一般和精炼集合), 实验结合亲和力以 pK_a ($-\log K_d$ 或者 $-\log K_i$) 值的形式收集自 PDBbind 网站 (<http://www.pdbbind.org.cn/>)。我们去除了 381 个无法通过 RDkit 处理的样本。其余的 12904 个样本随机分为训练 ($N = 11904$) 和验证 ($N = 1000$) 集合。三个独立的外部测试集合, 即 PDBbind 2013 核心集 ($N = 107$), 2016 核心集 ($N = 285$) 和 2019保留集 ($N = 4366$), 用于测试 GIGN 的泛化能力。2013 和 2016 核心集是评估结合亲和力预测性能的两个常用基准。^{3,5,10,14,35} 然而, 它们的小样本量往往导致过于乐观的结果。⁴因此, 我们从 PDBbind 2019版 收集了 4366 个样本。

在其他四个数据集中不可用的作为新的外部保留集, 模拟了一种真实的时间分割场景, 在该场景中, 对新发布的结构的结合亲和力通过在过去的结构数据上训练的模型进行预测。这种时间分割设置是一项更困难的任务, 并且更准确地反映了药物发现的真实情况。⁴⁴ 注意, 这五个数据集之间的复合物没有重叠。为了证明 CSAR-HiQ能够推广到非PDBbind数据集, 我们还在 CSAR-HiQ数据集中对其进行了验证。⁴⁵ CSAR-HiQ数据集包含两个子集, 总共有343个复合物。在剔除与上述五个数据集重叠并不能被RDkit处理的复合物后, 我们保留了47个样本作为非PDBbind外部测试集。在处理过程中, 距离大于5 Å的配体–蛋白质原子对被移除, 这遵循了之前的研究。^{11,33}

性能指标

模型性能通过皮尔逊相关系数 (R_p) 和均方根误差 (RMSE) 进行评估, 其定义如下。

$$\text{RMSE} = \sqrt{\frac{1}{|\mathcal{D}|} \sum_{n=1}^{|\mathcal{D}|} (\hat{y}_n - y_n)^2} \quad (14)$$

$$R_p = \frac{\sum_{i=1}^{|\mathcal{D}|} (\hat{y}_n - \bar{\hat{y}})(y_n - \bar{y})}{\sqrt{\sum_{i=1}^{|\mathcal{D}|} (\hat{y}_n - \bar{\hat{y}})^2 (y_n - \bar{y})^2}} \quad (15)$$

基线

我们将GIGN与以下代表性方法进行比较。

- 无交互方法:
 - DeepDTA¹⁷ 将配体和蛋白质分别表示为 SMILES和序列, 并建立两个卷积神经网络从 SMILES和序列中提取特征。
 - GraphDTA¹⁸ 与DeepDTA相似, 但将配体表示为二维图, 并使用图神经网络 (包括 GCN、GAT、GIN和GAT-GCN) 从图中学习表示。
 - MGraphDTA² 也将配体表示为二维图, 并使用多尺度图神经网络从图中学习表示。
- 基于交互的方法:
 - Pafnucy⁵ 是一个代表性的三维卷积神经网络模型, 它能够学习蛋白质–配体复合物的空间结构和物理交互。
 - OnionNet¹⁴ 基于蛋白质–配体交互壳中的接触数量生成基于交互的特征, 并使用二维卷积神经网络学习表示以进行预测。
 - PotentialNet¹⁵ 提出了一种分阶段的门控图神经网络, 以同时包含配体和蛋白质之间的分子内交互和非共价交互。
 - GNN-DTI¹¹ 是一种基于GNN的方法, 它结合了三维结构信息, 并使用距离感知的图注意力来区分每个相互作用对结合亲和力的贡献。

Table 1. Comparison Results of the Proposed GIGN and Baselines on Three External Test Sets

model		2013 core set (N = 107)		2016 core set (N = 285)		2019 holdout set (N = 4366)	
		RMSE	R _p	RMSE	R _p	RMSE	R _p
interaction-free methods	DeepDTA	1.639 (0.026)	0.718 (0.014)	1.357 (0.015)	0.785 (0.007)	1.485 (0.023)	0.586 (0.012)
	GCN	1.749 (0.062)	0.662 (0.032)	1.513 (0.048)	0.719 (0.023)	1.763 (0.039)	0.439 (0.021)
	GAT	2.043 (0.029)	0.476 (0.022)	1.748 (0.019)	0.594 (0.010)	1.663 (0.027)	0.432 (0.016)
	GIGN	1.691 (0.124)	0.694 (0.059)	1.470 (0.065)	0.743 (0.027)	1.676 (0.032)	0.472 (0.021)
	GAT-GCN	1.645 (0.085)	0.711 (0.036)	1.434 (0.064)	0.754 (0.025)	1.705 (0.075)	0.474 (0.028)
	MGraphDTA	1.680 (0.093)	0.696 (0.046)	1.439 (0.047)	0.753 (0.022)	1.553 (0.028)	0.538 (0.013)
interaction-based methods	Pafnucy	1.517 (0.014)	0.783 (0.005)	1.450 (0.047)	0.769 (0.019)	1.438 (0.016)	0.612 (0.014)
	OnionNet	1.583 (0.079)	0.741 (0.037)	1.399 (0.076)	0.770 (0.027)	1.510 (0.034)	0.573 (0.014)
	PotentialNet	1.607 (0.027)	0.773 (0.010)	1.503 (0.033)	0.772 (0.007)	1.514 (0.028)	0.564 (0.014)
	GNN-DTI	1.533 (0.084)	0.767 (0.040)	1.384 (0.013)	0.779 (0.008)	1.446 (0.006)	0.614 (0.007)
	IGN	1.428 (0.020)	0.807 (0.001)	1.269 (0.030)	0.821 (0.013)	1.410 (0.015)	0.630 (0.008)
	SchNet	1.570 (0.029)	0.754 (0.030)	1.390 (0.023)	0.787 (0.016)	1.522 (0.071)	0.560 (0.028)
	EGNN	1.498 (0.025)	0.782 (0.015)	1.289 (0.021)	0.816 (0.011)	1.399 (0.013)	0.628 (0.010)
	GIGN (ours)	1.380 (0.009)	0.821 (0.003)	1.190 (0.017)	0.840 (0.007)	1.393 (0.007)	0.641 (0.006)

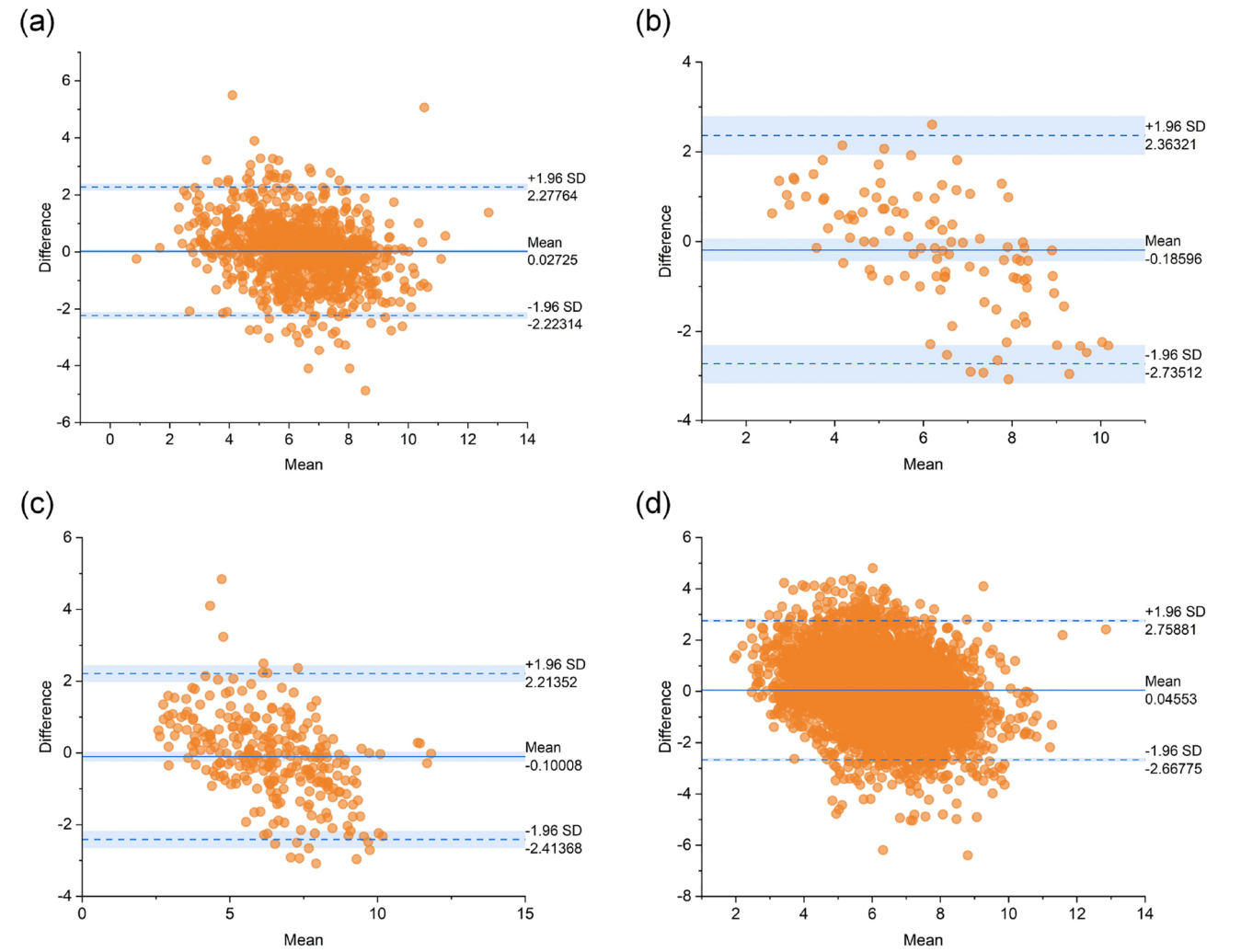


Figure 4. Bland–Altman plots for comparing predicted and experimental binding affinities for GIGN on the (a) validation set, (b) 2013 core set, (c) 2016 core set, and (d) 2019 holdout set, where the x -axis represents the mean (i.e., $(\hat{y}_n + y_n)/2$) and the y -axis represents the difference (i.e., $\hat{y}_n - y_n$).

• IGN³ designs two stacked GNN modules to sequentially learn the intramolecular and inter-molecular interactions, respectively.

• SchNet⁴² uses continuous filter convolutional layers to model local correlations, and it is invariant to translations and rotations.

表1. 提议的 GIGN 与三个外部测试集的基线比较结果

模型		2013 核心集 (N = 107)		2016 核心集 (N = 285)		2019保留集 (N = 4366)	
		均方根误差	R _p	均方根误差	R _p	均方根误差	R _p
无交互方法	DeepDTA	1.639 (0.026)	0.718 (0.014)	1.357 (0.015)	0.785 (0.007)	1.485 (0.023)	0.586 (0.012)
	GCN	1.749 (0.062)	0.662 (0.032)	1.513 (0.048)	0.719 (0.023)	1.763 (0.039)	0.439 (0.021)
	GAT	2.043 (0.029)	0.476 (0.022)	1.748 (0.019)	0.594 (0.010)	1.663 (0.027)	0.432 (0.016)
	图同质网络	1.691 (0.124)	0.694 (0.059)	1.470 (0.065)	0.743 (0.027)	1.676 (0.032)	0.472 (0.021)
	图注意力-图卷积网络	1.645 (0.085)	0.711 (0.036)	1.434 (0.064)	0.754 (0.025)	1.705 (0.075)	0.474 (0.028)
	MGraphDTA	1.680 (0.093)	0.696 (0.046)	1.439 (0.047)	0.753 (0.022)	1.553 (0.028)	0.538 (0.013)
基于交互的方法	Pafnucy	1.517 (0.014)	0.783 (0.005)	1.450 (0.047)	0.769 (0.019)	1.438 (0.016)	0.612 (0.014)
	OnionNet	1.583 (0.079)	0.741 (0.037)	1.399 (0.076)	0.770 (0.027)	1.510 (0.034)	0.573 (0.014)
	PotentialNet	1.607 (0.027)	0.773 (0.010)	1.503 (0.033)	0.772 (0.007)	1.514 (0.028)	0.564 (0.014)
	GNN-DTI	1.533 (0.084)	0.767 (0.040)	1.384 (0.013)	0.779 (0.008)	1.446 (0.006)	0.614 (0.007)
	IGN	1.428 (0.020)	0.807 (0.001)	1.269 (0.030)	0.821 (0.013)	1.410 (0.015)	0.630 (0.008)
	SchNet	1.570 (0.029)	0.754 (0.030)	1.390 (0.023)	0.787 (0.016)	1.522 (0.071)	0.560 (0.028)
	EGNN	1.498 (0.025)	0.782 (0.015)	1.289 (0.021)	0.816 (0.011)	1.399 (0.013)	0.628 (0.010)
	GIGN (我们的)	1.380 (0.009)	0.821 (0.003)	1.190 (0.017)	0.840 (0.007)	1.393 (0.007)	0.641 (0.006)

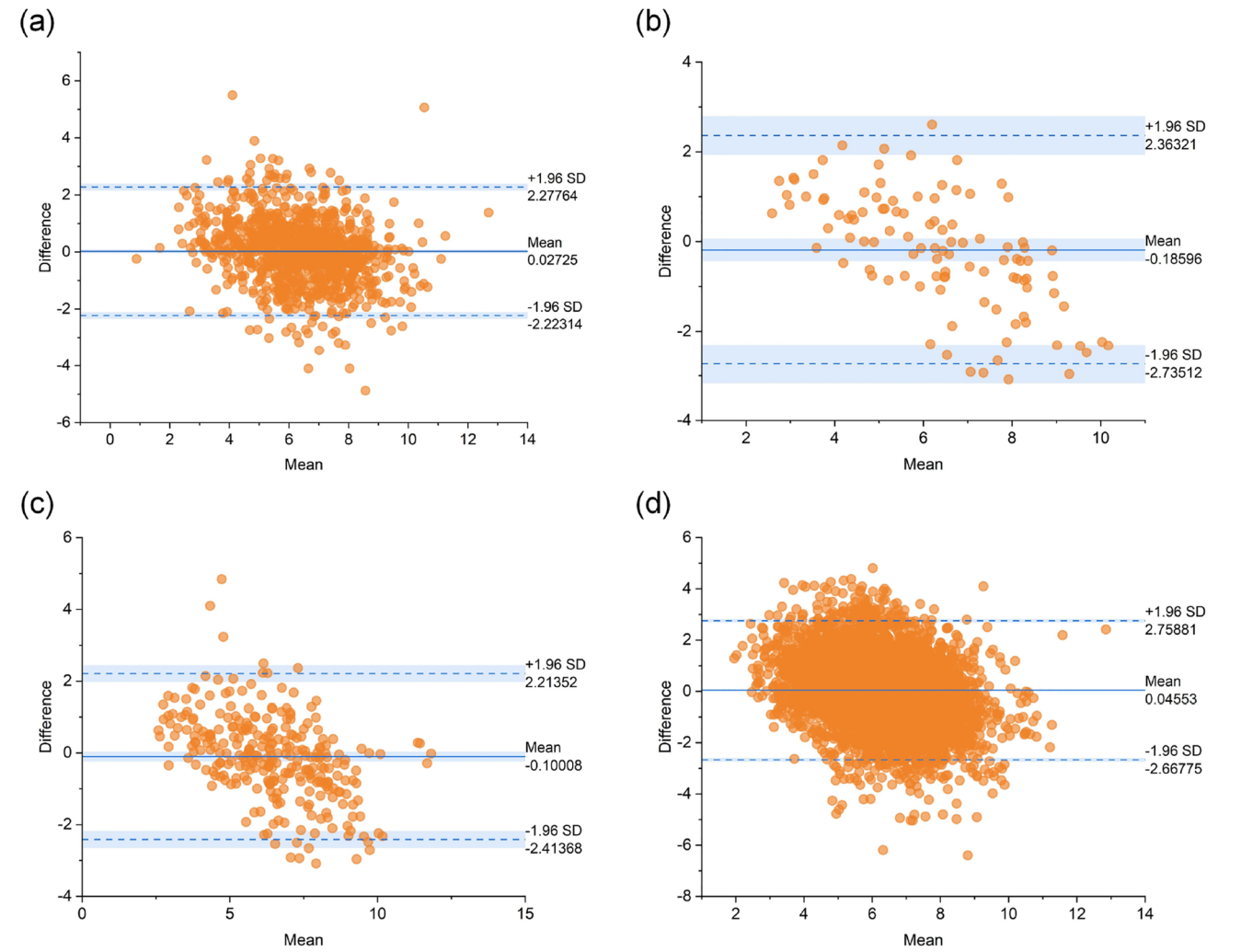


图4. 用于比较 GIGN 在 (a) 验证集、(b) 2013 核心集、(c) 2016 核心集和 (d) 2019 保留集上预测和实验结合亲和力的 Bland–Altman 图，其中 x 轴表示均值 (即, $(\hat{y}_n + y_n)/2$) 和 y 轴表示差值 (即, $\hat{y}_n - y_n$)。

•IGN³ 设计了两个堆叠的GNN模块，以分别顺序学习分子内和分子间相互作用。

•SchNet⁴² 使用连续滤波器卷积层来建模局部相关性，并且对平移和旋转不变。

Table 2. Experimental Results of the Ablation Study on Three External Test Sets

model	2013 core set		2016 core set		2019 holdout set	
	RMSE	R_p	RMSE	R_p	RMSE	R_p
GIGN_TMP	1.460 (0.008)	0.791 (0.003)	1.242 (0.015)	0.822 (0.004)	1.409 (0.004)	0.634 (0.004)
GIGN_TRVar	1.518 (0.047)	0.763 (0.019)	1.416 (0.024)	0.760 (0.010)	1.530 (0.004)	0.550 (0.007)
GIGN_RVar	1.520 (0.010)	0.765 (0.007)	1.344 (0.033)	0.788 (0.013)	1.447 (0.023)	0.608 (0.006)
GIGN (ours)	1.380 (0.009)	0.821 (0.003)	1.190 (0.017)	0.840 (0.007)	1.393 (0.007)	0.641 (0.006)

- EGNN³⁹ achieves state-of-the-art performance by exploring 3D translation and rotation symmetries.

All the baselines are implemented using the source code provided by the original papers. Note that GNN-DTI, SchNet, and EGNN were originally not designed for PLA predictions. Therefore, we tailor them to fit in with the PLA prediction task.

OVERALL COMPARISON

All the models share the same training, validation, and external test sets. All experiments are repeated three times, each with a different random seed. The validation set is used to select the best model with the lowest validation RMSE. The selected best model is used to evaluate the PLA prediction performance on three external test sets. We finally report the mean and standard deviation of performances based on three independent runs.

As shown in Table 1, it is observed that GIGN achieves the best performance on three external data sets, with 1.7%, 2.3%, and 1.7% improvements over the best baseline model in terms of R_p on the validation set, the 2013 core set, the 2016 core set, and the 2019 holdout set, respectively. Figure 4 shows the Bland–Altman plots for GIGN in the validation set, the 2013 core set, the 2016 core set, and the 2019 holdout set, where the differences between the predicted and experimental binding affinities are plotted against the averages of the affinities. The mean difference is the estimated bias, and the standard deviation (SD) of the differences measures the random fluctuations around this mean. It is observed that the mean differences for the four data sets are all around zero, which means that the predicted binding affinity is generally less biased. There are 94.60%, 95.37%, 95.08%, and 94.71% points within 95% limits of agreement (mean difference ± 1.96 standard deviation of the difference) for the four sets, which indicates good agreement between the predicted and experimental binding affinities. Moreover, we have the following observations.

As shown in Table 1, among the interaction-free methods, GraphDTA and MGraphDTA show worse performance than DeepDTA. However, these two 2D graph-based methods have shown superior performance compared to SMILES-based methods (e.g., DeepDTA) in Davis ($N = 30056$) and KIBA ($N = 118254$) data sets,^{2,17} which are much larger than PDBbind ver. 2016 ($N = 13285$) in terms of sample size. This result suggests that 2D graph-based methods easily overfit the training data when the sample size is relatively small. On the other hand, we can observe that models that incorporate 3D structure and physical interactions (i.e., PotentialNet, GNN-DTI, IGN, and GIGN) are superior to those 2D graph-based methods. This result suggests that, when data are scarce, it is necessary to inject the structure and interaction inductive biases into ML models to avoid learning these priors from insufficient amounts of data.

It is worth noting that all the methods only show a moderate generalization ability on the more difficult PDBind ver. 2019 holdout set. This result suggests that accurately predicting PLA is still a challenge. Given that a threshold value of 0.600 for R_p is

typically used to qualify a good predictive quantitative structure–activity relationship model,^{35,46} only Pafnucy, GNN-DTI, IGN, EGNN, and GIGN can be considered satisfactory.

To explore why models achieve worse performance in the 2019 holdout set, we calculate the protein overlap rate and the ligand overlap rate between the test set and the training set. The overlap rate indicates the percentage of proteins (or ligands) in the test set that can also be found in the training set. The protein overlap rates are 100.0%, 99.6%, and 68.9% for the 2013 test set, the 2016 test set, and the 2019 holdout set, respectively. The ligand overlap rates are 37.4%, 30.9%, and 24.8% for the three sets. Obviously, the protein and ligand overlap rates for the 2013 test set and the 2016 test set are much higher than those for the 2019 holdout set, which can explain why the models achieve inferior performance in the 2019 holdout set. Although the complexes in the 2013 and 2016 test sets are completely different from the training set, the high overlap rate of proteins and ligands may cause information leakage, thus leading to overoptimistic results.

ABLATION STUDY

To verify the effectiveness of the proposed heterogeneous interaction layer and the invariance constraint, we compare GIGN with its three variants on the three benchmarks.

- GIGN_TMP uses traditional message passing where covalent and noncovalent interactions are considered the same type of interaction.
- GIGN_VarTR is variant to translations and rotations. Specifically, we add the corresponding coordinate r_i ($\times 0.1$ to stabilize training) to the initial feature x_i for node v_i (i.e., $(r_i \times 0.1) \parallel x_i$, where $\parallel$ is concatenation) to inject structural information to the initial node features. Obviously, these coordinates are variant to translations and rotations and thus $(r_i \times 0.1) \parallel x_i$ is also variant to the transformations.
- GIGN_VarR is only variant to rotations. The coordinates of the complex are normalized by subtracting the centroid of the ligand before adding them to the initial feature x_i .

Table 2 summarizes the experimental results of the ablation study. It is observed that GIGN exceeds the three variants, demonstrating the necessity of the heterogeneous interaction setting and the invariance constraint. Specifically, GIGN_VarR surpasses GIGN_VarTR greatly in the 2016 core set and the 2019 holdout set, which proves the importance of translational invariance. Without translational invariance, the model is less able to discern the same protein–ligand complex in different positions (i.e., it predicts different binding affinities for the same complexes in different positions). GIGN_VarR solves this problem by normalizing the coordinates, leading to a better generalization capability. Moreover, it can be observed that GIGN outperforms GIGN_VarR by a large margin, indicating the significance of rotational invariance. Another way to mitigate

表2. 三个外部测试集的消融研究的实验结果

模型	2013 核心集		2016 核心集		2019保留集	
	均方根误差	R_p	均方根误差	R_p	均方根误差	R_p
GIGN_TMP	1.460 (0.008)	0.791 (0.003)	1.242 (0.015)	0.822 (0.004)	1.409 (0.004)	0.634 (0.004)
GIGN_TRVar	1.518 (0.047)	0.763 (0.019)	1.416 (0.024)	0.760 (0.010)	1.530 (0.004)	0.550 (0.007)
GIGN_RVar	1.520 (0.010)	0.765 (0.007)	1.344 (0.033)	0.788 (0.013)	1.447 (0.023)	0.608 (0.006)
GIGN (我们的)	1.380 (0.009)	0.821 (0.003)	1.190 (0.017)	0.840 (0.007)	1.393 (0.007)	0.641 (0.006)

- EGNN³⁹ 通过探索三维平移和旋转对称性来实现最先进性能。

所有基线模型均使用原始论文提供的源代码进行实现。请注意，GNN-DTI、SchNet 和 EGNN 最初并未针对 PLA 预测而设计。因此，我们对它们进行了调整，以适应 PLA 预测任务。

总体比较

所有模型共用相同的训练、验证和外部测试集。所有实验重复三次，每次使用不同的随机种子。验证集用于选择验证均方根误差（RMSE）最低的最佳模型。所选的最佳模型用于评估在三个外部测试集上的 PLA 预测性能。最终，我们根据三次独立运行报告性能的均值和标准差。

如表1所示，在三个外部数据集上观察到 GIGN 的性能最佳，分别在验证集、2013 核心集、2016 核心集和 2019 保留集上，相比最佳基线模型提高了 1.7%、2.3% 和 1.7% 的 R_p 。图 4 展示了 GIGN 在验证集、2013 核心集、2016 核心集和 2019 保留集上的布兰德-阿尔特曼图，其中预测结合亲和力与实验结合亲和力之间的差异相对于亲和力的平均值进行绘制。均值差为估计偏差，差异的标准差（SD）则衡量了该均值周围的随机波动。观察到四个数据集的均值差均接近零，这表明预测结合亲和力通常偏差较小。四个数据集中，分别有 94.60%、95.37%、95.08% 和 94.71% 的点落在 95% 协议限内（均值差 ± 1.96 标准差）之内，这表明预测与实验结合亲和力之间存在良好一致性。此外，我们有以下观察。

如表1所示，在无交互方法中，GraphDTA 和 MGraphDTA 的表现优于 DeepDTA。然而，这两种基于2D图的方法在 Davis ($N = 30056$) 和 KIBA ($N = 118254$) 数据集上相较于基于 SMILES 的方法（例如，DeepDTA）表现较差，^{2,17} 后者在样本规模上远大于 PDBbind 2016版 ($N = 13285$)。这一结果表明，当样本规模相对较小时，基于2D图的方法容易过拟合训练数据。另一方面，我们可以观察到，结合3D结构和物理交互的模型（即，PotentialNet、GNN-DTI、IGN 和 GIGN）优于那些基于2D图的方法。这一结果表明，当数据稀缺时，有必要将结构和交互的归纳偏见注入到机器学习模型中，以避免从不足量的数据中学习这些先验知识。

值得注意的是，所有方法在更具挑战性的 PDBind 2019版本保留集上仅显示出适度的泛化能力。这个结果表明，准确预测PLA仍然是一项挑战。考虑到 R_p 的阈值为0.600，

通常用于确定一个好的预测定量结构–活性关系模型的^{35,46}，仅有Pafnucy、GNN-DTI、IGN、EGNN和 GIGN可以被视为令人满意。

为探讨为何模型在2019保留集中的表现较差，我们计算了测试集与训练集之间的蛋白质重叠率和配体重叠率。重叠率表示在测试集中可以在训练集中找到的蛋白质（或配体）的百分比。2013测试集、2016测试集和2019保留集的蛋白质重叠率分别为100.0%、99.6%和68.9%。这三个数据集的配体重叠率分别为37.4%、30.9%和24.8%。显然，2013测试集和2016测试集的蛋白质和配体重叠率远高于2019保留集，这可以解释为何模型在2019保留集中的表现较差。尽管2013和2016测试集中的复合物与训练集完全不同，但高的蛋白质和配体重叠率可能导致信息泄露，从而导致过于乐观的结果。

消融研究

为了验证所提出的异质交互层和不变性约束的有效性，我们在三个基准上比较GIGN及其三个变体。

- GIGN_TMP采用传统的消息传递，其中共价交互和非共价交互被视为同一种交互类型。

- GIGN_VarTR对平移和旋转具有变体。具体而言，我们将相应的坐标 r_i ($\times 0.1$ 以稳定训练)添加到节点 v_i 的初始特征 x_i 中（即， $(r_i \times 0.1) \parallel x_i$ ，其中 $\parallel$ 是连接）以在初始节点特征中注入结构信息。显然，这些坐标对平移和旋转具有变体，因此 $(r_i \times 0.1) \parallel x_i$ 也对这些变换具有变体。

- GIGN_VarR仅对旋转具有变体。络合物的坐标通过减去配体的质心进行归一化，然后再将其添加到初始特征 x_i 中。

表2总结了消融研究的实验结果。观察到GIGN超越了三个变体，展示了异质交互设置和不变性约束的必要性。具体而言，GIGN_VarR在2016核心集和2019保留集中显著超过GIGN_VarTR，这证明了平移不变性的重要性。如果没有平移不变性，该模型不太能够辨别同一蛋白质–配体复合物在不同位置的表现（即，它为不同位置的相同复合物预测不同的结合亲和力）。GIGN_VarR通过规范化坐标解决了这个问题，从而带来了更好的泛化能力。此外，可以观察到GIGN大幅超过GIGN_VarR，这表明了旋转不变性的意义。另一种缓解

rotational variance is to use data augmentation (i.e., randomly rotating the input complex during training), which is the strategy that Pafnucy adopts. GIGN enforces the model to recognize the same protein–ligand complex with different initial placements via invariance constraint, thus avoiding expensive data augmentation strategies.

■ PERFORMANCE EVALUATION ON THE NON-PDBBIND EXTERNAL TEST SET

To further evaluate the generalization ability of the IGNNs, we also evaluate them on the CSAR-HiQ data set. As shown in Table 3, IGN, SchNet, EGNN, and GIGN achieve promising

Table 3. Performance Evaluation for Six IGNNs on the CSAR-HiQ Dataset

model	RMSE	R_p
PotentialNet	1.761 (0.073)	0.647 (0.037)
GNN-DTI	1.740 (0.094)	0.657 (0.049)
IGN	1.618 (0.130)	0.714 (0.057)
SchNet	1.587 (0.077)	0.746 (0.016)
EGNN	1.538 (0.010)	0.760 (0.012)
GIGN (ours)	1.498 (0.058)	0.774 (0.025)

performance in this non-PDBbind external test set, which suggests that the high performance on the PDBbind data set may be translated into high performance in real-world virtual screening and drug development.

■ GIGN RETAINS ITS SUPERIORITY IN REDOCKED STRUCTURES

There is a risk that a model trained only with the experimental structures cannot be generalized to redocked structures¹⁴ (i.e., the robustness of the model may be questionable). To assess the robustness of GIGN, we also evaluate it using redocked structures. Specifically, for each complex in the 2016 test set, we select a redocked structure with the lowest RMSD (RMSD < 2 Å, also called native-like pose) from the CASF-2016 benchmark.⁴⁷ We use GIGN to predict the binding affinities of these redocked structures. The R_p and RMSE values for the predicted binding affinities of redocked structures against the experimental binding affinity are 0.826 (0.005) and 1.261 (0.014), respectively. These results suggest that the predicted binding affinities for redocked structures are also close to experimental binding affinities.

■ PERFORMANCE EVALUATION UNDER THE COLD START SETTING

To further prevent the structural information on proteins or ligands from leaking into the test set, we split the original

training set ($N = 11904$) into new training, validation, and test sets in a ratio of 8:1:1 according to 30% protein sequence identity or ligand scaffold (also called cold start setting²). We use the random split setting as a control group. It is not surprising that the models achieve much worse performance in the cold start setting, as shown in Table 4, because the cold start setting prevents the structural information on proteins or ligands from leaking into the test set. Moreover, it can be observed that GIGN achieves better performance in the random and scaffold split settings, while it obtains relatively inferior performance in the sequence identity split setting. This is because the heterogeneous interaction layer of GIGN is mainly designed to enhance the node representations of ligand atoms instead of protein atoms.

■ EPISTEMIC UNCERTAINTY QUANTIFICATION OF GIGN

ML-based methods may produce unreliable predictions for those out-of-domain complexes. One possible way to reduce unreliable predictions is to filter predictions according to their estimated uncertainties.⁸ We use Monte Carlo dropout (MCDO) to quantify epistemic uncertainties⁴⁸ of the predictions. In particular, we estimate the uncertainties of predictions for the 2019 holdout set, which has a relatively large sample size. The predictions are first ranked according to the estimated uncertainties. We then select one-third of the samples with the lowest uncertainties (low group) and one-third of the samples with the highest uncertainties (high group). As a comparison, we randomly select one-third of the samples from the 2019 holdout set and label it as a random group. A baseline group that simply uses all samples in the data set is also provided for a better illustration. Figure 5(a) shows the comparison between the four groups in terms of R_p . We can observe that the R_p of the low group is much larger than those of the random and low groups, which suggests the estimated uncertainties can be used to filter unreliable predictions.

To figure out why ML models make some predictions with high uncertainty, we first plot the histogram of experimental binding affinities for the training set and the scatter diagram between the experimental binding affinity and the estimated uncertainty for the 2019 holdout set, as shown in Figure 5(b) and (c). It is observed that the experimental binding affinities generally follow a normal distribution, as shown in Figure 5(b). Intuitively, the model would show high uncertainty for those complexes with low/high experimental binding affinities due to the smaller sample size. However, the result illustrated in Figure 5(c) reveals that smaller sample size is not the main reason for the high uncertainties of the predictions because the estimated uncertainties for those complexes with low/high experimental binding affinities are not much higher than others. A possible

Table 4. Performance evaluation for six IGNNs in the random split and cold start (split based on sequence identity and scaffold) scenarios

model	random		sequence identity (30%)		scaffold	
	RMSE	R_p	RMSE	R_p	RMSE	R_p
PotentialNet	1.360 (0.030)	0.691 (0.013)	1.582 (0.017)	0.507 (0.006)	1.538 (0.006)	0.532 (0.015)
GNN-DTI	1.403 (0.011)	0.666 (0.007)	1.499 (0.008)	0.562 (0.006)	1.401 (0.027)	0.631 (0.018)
IGN	1.295 (0.011)	0.732 (0.006)	1.509 (0.034)	0.561 (0.025)	1.361 (0.041)	0.665 (0.023)
SchNet	1.398 (0.044)	0.672 (0.021)	1.499 (0.022)	0.566 (0.020)	1.456 (0.046)	0.607 (0.014)
EGNN	1.285 (0.025)	0.730 (0.012)	1.483 (0.029)	0.579 (0.017)	1.395 (0.036)	0.635 (0.029)
GIGN (ours)	1.264 (0.015)	0.746 (0.006)	1.521 (0.012)	0.565 (0.007)	1.353 (0.002)	0.685 (0.004)

旋转不变性的方法是使用数据增强（即在训练期间随机旋转输入复合物），这是Pafnucy所采用的策略。GIGN通过不变性约束强制模型识别具有不同初始位置的相同蛋白质–配体复合物，从而避免昂贵的数据增强策略。

■ 在非-PDBbind外部测试集上的性能评估

为了进一步评估IGNNs的泛化能力，我们还在CSAR-HiQ数据集上对其进行评估。如表3所示，IGN、SchNet、EGNN和GIGN取得了可喜的

表3. 六种 IGNN 在 CSAR-HiQ 数据集上的性能评估

模型	均方根误差	R_p
PotentialNet	1.761 (0.073)	0.647 (0.037)
GNN-DTI	1.740 (0.094)	0.657 (0.049)
IGN	1.618 (0.130)	0.714 (0.057)
SchNet	1.587 (0.077)	0.746 (0.016)
EGNN	1.538 (0.010)	0.760 (0.012)
GIGN（我们的）	1.498 (0.058)	0.774 (0.025)

在此非-PDBbind外部测试集中的表现，这表明在PDBbind数据集上的高性能可能会转化为在实际虚拟筛选和药物开发中的高性能。

■ GIGN在重dock结构中保持其优越性

存在一个风险，即仅使用实验结构训练的模型无法推广到重dock结构¹⁴（即，模型的鲁棒性可能存在疑问）。为了评估GIGN的鲁棒性，我们还使用重dock结构对其进行评估。具体而言，对于2016测试集中的每一个复合体，我们从CASF-2016基准中选择具有最低RMSD（RMSD <2 Å，也称为本征类似构象）的重dock结构。⁴⁷ 我们使用GIGN预测这些重dock结构的结合亲和力。对于重dock结构的预测结合亲和力与实验结合亲和力的 R_p 和均方根误差（RMSE）值分别为0.826（0.005）和1.261（0.014）。这些结果表明，重dock结构的预测结合亲和力也接近实验结合亲和力。

■ 在冷启动设置下的性能评估

为了进一步防止蛋白质或配体的结构信息泄漏到测试集中，我们分割了原始

表4. 在随机划分和冷启动（基于序列相似性和支架的划分）场景中对六个IGNN的性能评估

模型	随机		序列相似性（30%）		支架	
	均方根误差	R_p	均方根误差	R_p	均方根误差	R_p
PotentialNet	1.360 (0.030)	0.691 (0.013)	1.582 (0.017)	0.507 (0.006)	1.538 (0.006)	0.532 (0.015)
GNN-DTI	1.403 (0.011)	0.666 (0.007)	1.499 (0.008)	0.562 (0.006)	1.401 (0.027)	0.631 (0.018)
IGN	1.295 (0.011)	0.732 (0.006)	1.509 (0.034)	0.561 (0.025)	1.361 (0.041)	0.665 (0.023)
SchNet	1.398 (0.044)	0.672 (0.021)	1.499 (0.022)	0.566 (0.020)	1.456 (0.046)	0.607 (0.014)
EGNN	1.285 (0.025)	0.730 (0.012)	1.483 (0.029)	0.579 (0.017)	1.395 (0.036)	0.635 (0.029)
GIGN (我们的)	1.264 (0.015)	0.746 (0.006)	1.521 (0.012)	0.565 (0.007)	1.353 (0.002)	0.685 (0.004)

训练集 ($N = 11904$) 按照 30% 蛋白质序列相似性或配体支架的比例分成新的训练集、验证集和测试集，比例为 8:1:1（也称为冷启动设置²）。我们使用随机划分设置作为对照组。在冷启动设置中模型的性能远不及其他设置，这并不令人惊讶，如表4所示，因为冷启动设置防止了蛋白质或配体的结构信息泄漏到测试集中。此外，可以观察到 GIGN 在随机和支架划分设置中取得了更好的性能，而在序列相似性划分设置中表现相对较差。这是因为 GIGN 的异质交互层主要旨在增强配体原子的节点表示，而非蛋白质原子的节点表示。

■ GIGN 的认知不确定性量化

基于ML的方法在那些域外复合物上可能产生不可靠的预测。减少不可靠预测的一个可能方法是根据估计的不确定性过滤预测。⁸ 我们使用蒙特卡洛丢弃（MCDO）来量化预测的认知不确定性⁴⁸。具体而言，我们估计2019保留集的预测不确定性，该集样本量相对较大。首先根据估计的不确定性对预测进行排名。然后，我们选择不确定性最低的三分之一样本（低组）和不确定性最高的三分之一样本（高组）。作为比较，我们从2019保留集中随机选择三分之一样本，并将其标记为随机组。为了更好地说明，还提供了一个简单地使用数据集中所有样本的基准组。图5(a)显示了四组在 R_p 方面的比较。我们可以观察到低组的 R_p 远大于随机组和高组，这表明估计的不确定性可以用于过滤不可靠的预测。

为了弄清楚为什么机器学习模型会对某些预测产生较高的不确定性，我们首先绘制了训练集的实验结合亲和力的直方图，以及2019保留集中实验结合亲和力与估计不确定性之间的散点图，如图5(b)和(c)所示。观察到实验结合亲和力通常遵循正态分布，如图5(b)所示。直观上，模型对于那些低/高实验结合亲和力的复合物会表现出高不确定性，这是由于样本量较小。然而，图5(c)中所示的结果显示，样本量较小并不是预测高度不确定性的主要原因，因为那些低/高实验结合亲和力的复合物的估计不确定性并不会比其他复合物高太多。一个可能的

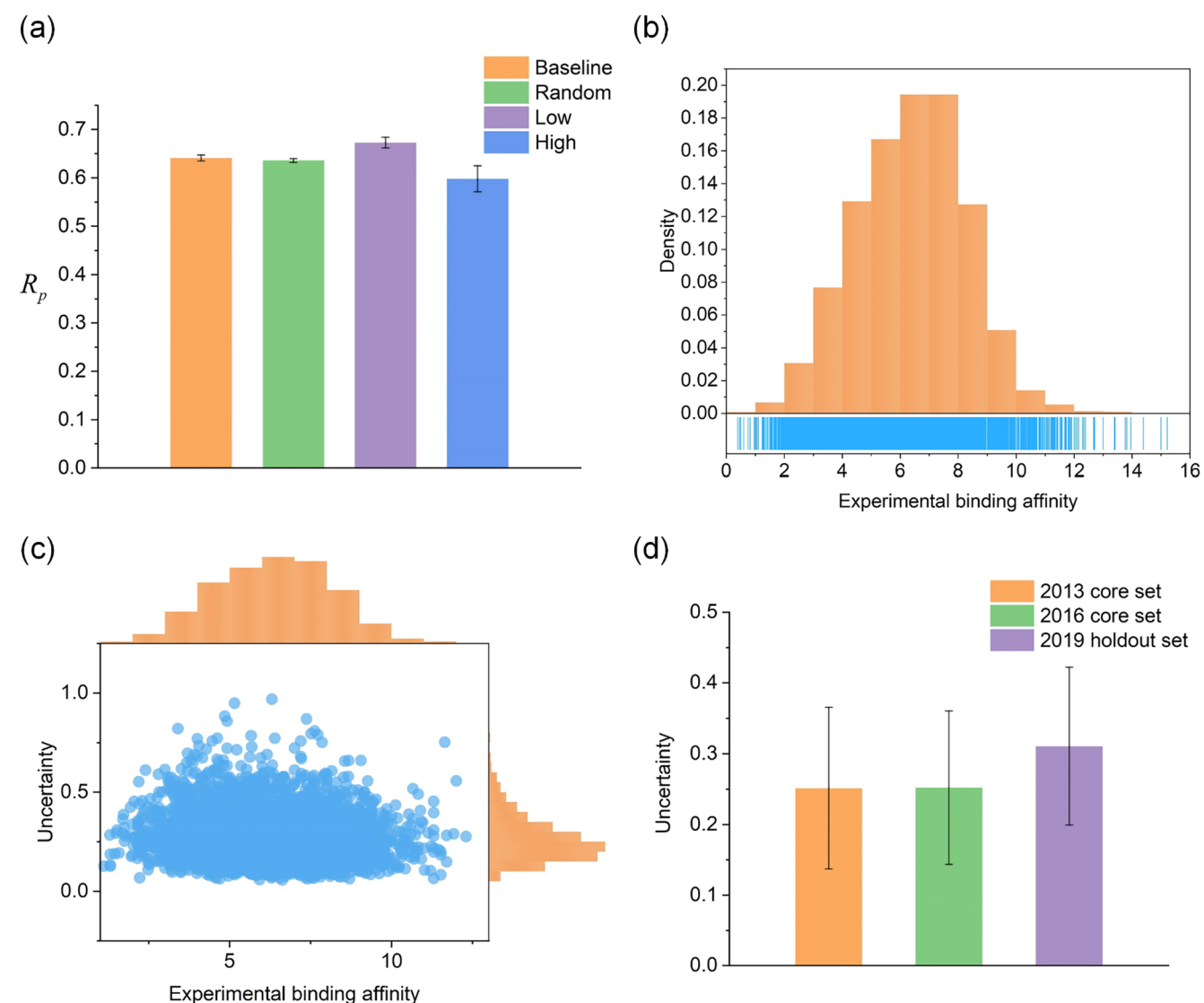


Figure 5. Uncertainty estimation of predictions for GIGN. (a) R_p values of baseline, random, low, and high groups on the 2019 holdout set (baseline, using the whole data set; random, randomly selecting one-third of the samples; low, the one-third of the samples with the lowest uncertainties; and high, the one-third of the samples with the highest uncertainties). (b) Histogram of experimental binding affinities for the training set. (c) Scatter diagram of uncertainty against experimental binding affinity for the 2019 holdout set. (d) Estimated uncertainties for predictions in the 2013 core set, the 2016 core set, and the 2019 holdout set.

reason for high uncertainties is that some inputs are out-of-domain complexes. Concretely, Figure 5(d) depicts the estimated uncertainties for predictions in the 2013 core set, the 2016 core set, and the 2019 holdout set (the distribution of the estimated uncertainty is shown in Supplement B Figure S1). It can be observed that the estimated uncertainties for the 2019 holdout set are much higher than those for the 2013 and 2016 core sets. The 2019 holdout set is collected following the temporal split setting, and it contains newly released structures that may exhibit different patterns (i.e., out-of-domain complexes) compared to complexes in the training set, which can explain why the estimated uncertainties for it are higher than those for the 2013 and 2016 core sets and why GIGN achieves relatively worse performance in this data set.

■ VISUAL EXPLANATIONS FOR GIGN

GNNs cannot be fully trusted without understanding and verifying their working mechanisms,^{43,49} which limits their

application in drug discovery scenarios. In this section, we conduct two visual explanation experiments to rationalize GIGN.

To demonstrate that the learned graph-level representations of complexes (i.e., h_G in eq 12) are meaningful, we visualize them at some stages during training. Specifically, we randomly select 256 complexes from PDBbind ver. 2016 and use GIGN to map them to m -dimensional vectors ($m = 256$ in our study). The 256-dimensional vectors, similar to the fingerprint representation of a complex, are reduced to 2D embeddings by t-SNE.⁵⁰ Figure 6 shows the distribution of 2D embeddings on epochs 1, 80, 175, and 302 (the best epoch with the lowest validation RMSE). As shown in Figure 6, the scatter plots show some degree of chaos at the beginning and then group into clusters during training according to the binding affinities, which suggests that the learned graph representations are meaningful.

To further explore which ligand atoms provide the most significant contribution to PLA prediction, we visualize the

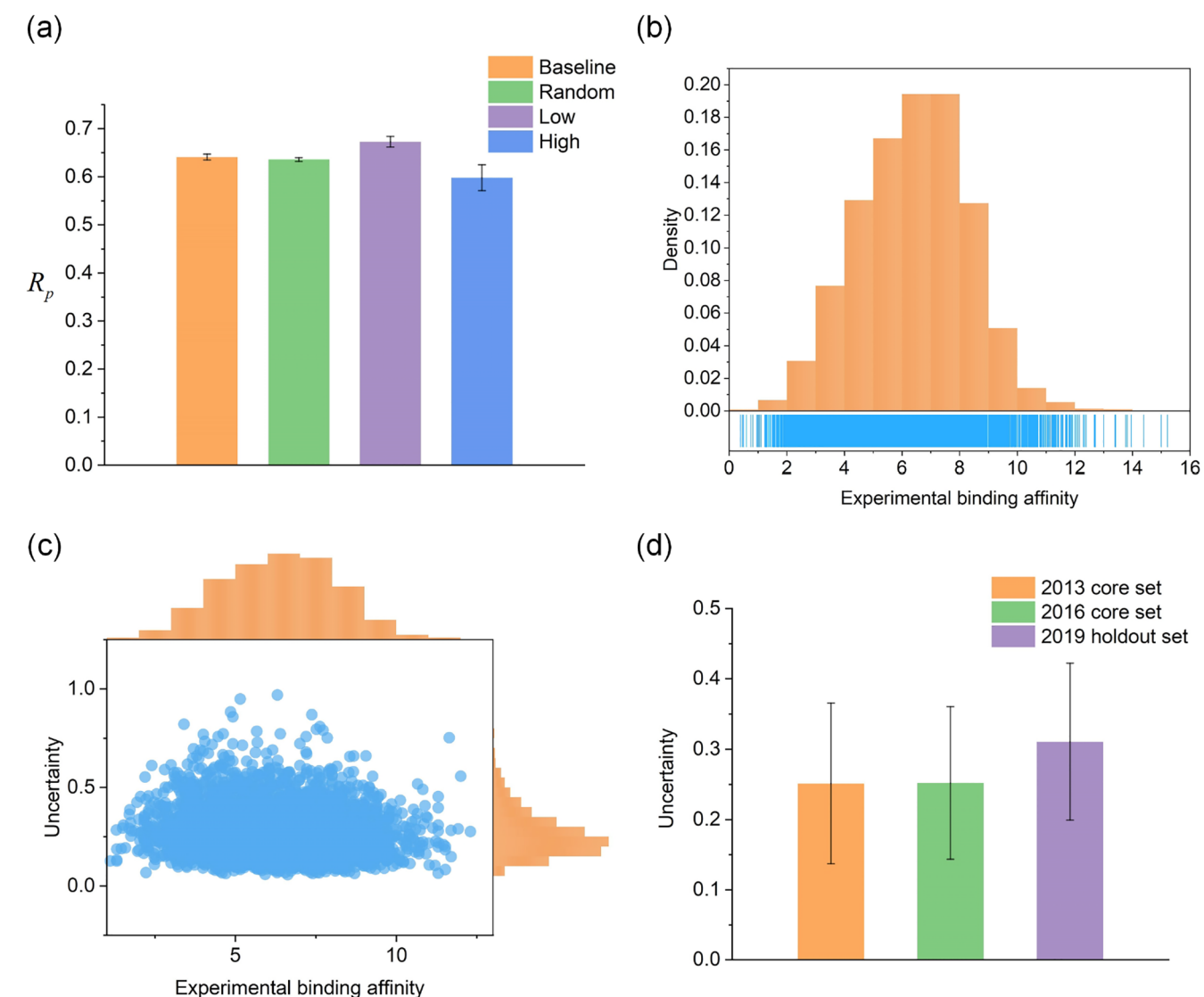


图5. GIGN预测的不确定性估计。(a) 2019保留集的基线、随机、低和高组的 R_p 值（基线，使用整个数据集；随机，随机选择三分之一的样本；低，具有最低不确定性的三分之一样本；高，具有最高不确定性的三分之一样本）。(b) 训练集的实验结合亲和力直方图。(c) 2019保留集的不确定性与实验结合亲和力的散点图。(d) 2013核心集、2016核心集和2019保留集的预测不确定性估计。

高不确定性的原因在于某些输入为域外复合物。具体而言，图5(d)描述了在2013核心集、2016核心集和2019保留集中的预测估计不确定性（估计不确定性的分布在补充材料B图S1中显示）。可以观察到，2019保留集的估计不确定性远高于2013和2016核心集。2019保留集是按照时间拆分设置收集的，且它包含的新发布结构可能与训练集中的复合物表现出不同模式（即，域外复合物），这可以解释其估计不确定性为何高于2013和2016核心集，以及为何GIGN在该数据集中表现相对较差。

■ GIGN的可视化解释

在没有理解和验证其工作机制的情况下，图神经网络（GNNs）无法得到完全信任，^{43,49}这限制了它们的

在药物发现场景中的应用。在本节中，我们进行两项可视化解释实验，以合理化GIGN。

为了证明复合物的图级表示（即在eq 12中）是有意义的，我们在训练的某些阶段对它们进行可视化。具体而言，我们从PDBbind 2016版中随机选择256个复合物，并使用GIGN将它们映射到 m 维向量（ $m = 256$ 在本研究中）。256维向量类似于复合物的指纹表示，通过t-SNE被降维为二维嵌入。⁵⁰图6显示了在第1、80、175和302个epoch的二维嵌入分布（具有最低验证均方根误差的最佳epoch）。如图6所示，散点图在开始时显示出一定程度的混乱，然后根据结合亲和力在训练过程中聚集成簇，这表明学习到的图表示是有意义的。

为了进一步探索哪些配体原子对PLA预测贡献最大，我们可视化了

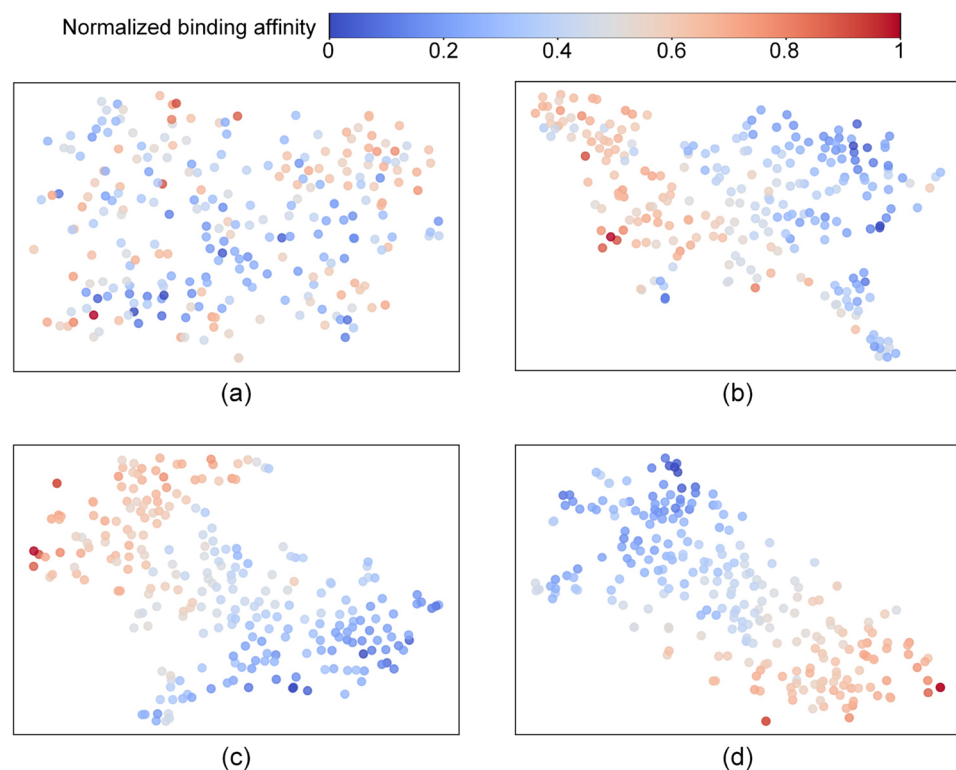


Figure 6. Visualizations of complex embeddings at epochs (a) 1, (b) 80, (c) 175, and (d) 302.

ligand atom importance using Grad-AAM.² Figure 7 shows the visualization results, where the distribution of atom importance shows a close match to the binding mode (more examples can be found in Supplement B Figure S2). Specifically, we have two following observations. First, the ligand atoms with higher weights tend to locate at positions with hydrogen bonds that are generally considered to be facilitators of protein–ligand binding. Second, those ligand atoms with lower weights are generally outside the binding pocket, which suggests that GIGN is sensitive to binding poses. We also provide a case about visualizing protein atom importance using Grad-AAM (Figure 8). Similarly, it can be observed that the atom importance has a relationship with hydrogen bonds. Overall, these results suggest that GIGN may be able to capture essential features relevant to binding.

PARAMETER SENSITIVITY ANALYSIS

In this experiment, two major hyper-parameters of GIGN are tested, which are the number of heterogeneous interaction layers T and the number of hidden channels m . We analyze the effect of T ranging from 1 to 6. As shown in Figure 9, the model achieves better performance when using 2–4 heterogeneous interaction layers. Theoretically, with the increase of T , the receptive field of GNN can be enlarged⁴³ (i.e., the model can detect larger molecular substructures). However, increasing T would also cause the oversmoothing problem.⁴³ Therefore, applying three heterogeneous layers for GIGN is already enough to capture sufficient structural information. We analyze the effect of m by setting it to different values (32, 64, 128, 192, 256, and 320). As shown in Figure 9, a general upward trend can be observed by increasing the value of m from 32 to 256. This is because the relatively larger hidden channels can encode more information. However, a slight degradation of performance can be found if we

continue to increase it to 320, which is probably caused by the overfitting problem.

MODEL COMPLEXITY

Besides the state-of-the-art performance on PLA prediction, another advantage of GIGN is the low computational cost. Figure 10 shows the number of parameters and floating point operations (FLOPs) for six IGNNs (PotentialNet, GNN-DTI, IGN, SchNet, EGNN, and GIGN). Among the top three IGNNs (IGN, EGNN, and GIGN) that have better performance than the others, as shown in Table 1, GIGN has a much lower computational cost than IGN and EGNN. This lightweight advantage implies that GIGN is more readily applicable to large-scale databases.

LIMITATIONS

One disadvantage of interaction-based models is that they heavily rely on optimized structures. Specifically, all complexes in the PDBbind data set are optimized according to the minimum energy criterion, which means that the ligand binding structure (i.e., the best position and orientation of the ligand) should be provided in order to train the interaction-based models. However, optimized structures may not always be available in real applications. Luckily, EQUIBIND¹ shows that it is possible to use deep learning to predict ligand binding structures with significant speed-ups and better quality than traditional docking. If the optimized structures are unknown, EQUIBIND can be a useful tool to predict how a ligand binds to the protein.

This work focuses on the study of how to effectively incorporate 3D structures of complexes and protein–ligand physical interactions into GNNs. To this end, we propose a simple but effective GNN-based model, GIGN, to predict the

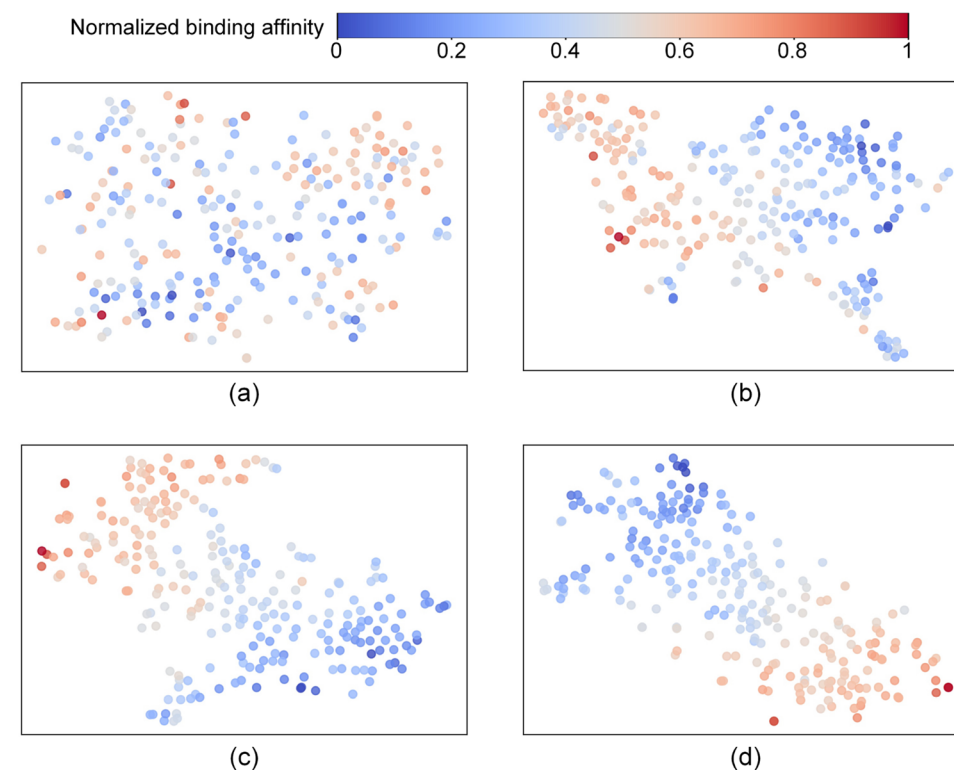


图6. 复合物嵌入在训练周期(a) 1, (b) 80, (c) 175, 以及(d) 302的可视化结果。

使用Grad-AAM的配体原子重要性。² 图7展示了可视化结果, 其中原子重要性的分布与结合模式密切匹配 (更多示例见补充材料B图S2)。具体而言, 我们有两个以下观察。首先, 权重较高的配体原子往往位于被普遍认为是促进蛋白质–配体结合的氢键的位置。其次, 那些权重较低的配体原子通常位于结合口袋之外, 这表明GIGN对结合姿势敏感。我们还提供了一个关于使用Grad-AAM可视化蛋白质原子重要性的案例 (图8)。同样, 可以观察到原子重要性与氢键之间的关系。总体而言, 这些结果表明GIGN能够捕捉与结合相关的基本特征。

参数敏感性分析

在本实验中, 测试了 GIGN 的两个主要超参数, 即异构交互层的数量 T 以及隐藏通道的数量 m 。我们分析了 T 从1到6的影响。如图9所示, 当使用2–4异构交互层时, 模型实现了更好的性能。从理论上讲, 随着 T 的增加, GNN的感受野可以扩大⁴³ (即模型可以检测到更大的分子亚结构)。然而, 增加 T 还会导致过平滑问题。⁴³ 因此, 为 GIGN 应用三个异构层已经足以捕捉到足够的结构信息。我们通过将 m 设置为不同的值 (32、64、128、192、256 和 320) 来分析其影响。如图9所示, 随着 m 从32增加, 可以观察到一个普遍的上升趋势。

256. 这是因为相对较大的隐藏通道可以编码更多信息。然而, 如果我们

继续将其增加到320, 则可能会出现轻微的性能下降, 这可能是由过拟合问题引起的。

模型复杂性

除了在 PLA 预测上表现最先进, GIGN 的另一个优势是低计算成本。

图10显示了六个IGNNs (PotentialNet、GNN-DTI、IGN、SchNet、EGNN和GIGN) 的参数数量和浮点运算数 (FLOPs)。在表现优于其他模型的前三个 IGNNs (IGN、EGNN和GIGN) 中, 如表1所示, GIGN的计算成本明显低于IGN和EGNN。这种轻量级优势意味着GIGN更易于应用于大规模数据库。

局限性

基于交互的模型的一个缺点是它们在很大程度上依赖于优化后的结构。具体而言, PDBbind数据集中所有复合物都是依据最小能量标准进行优化的, 这意味着在训练基于交互的模型时, 必须提供配体结合结构 (即配体的最佳位置和取向)。然而, 优化后的结构在实际应用中并不总是可用。幸运的是, EQUIBIND¹表明, 使用深度学习来预测配体结合结构是可能的, 并且比传统对接具有显著的速度提升和更好的质量。如果优化后的结构未知, EQUIBIND可以成为预测配体如何与蛋白质结合的有用工具。

本研究专注于如何有效地将复合物的三维结构和蛋白–配体的物理交互纳入图神经网络 (GNNs) 中。为此, 我们提出了一个简单但有效的基于GNN的模型 GIGN, 以预测

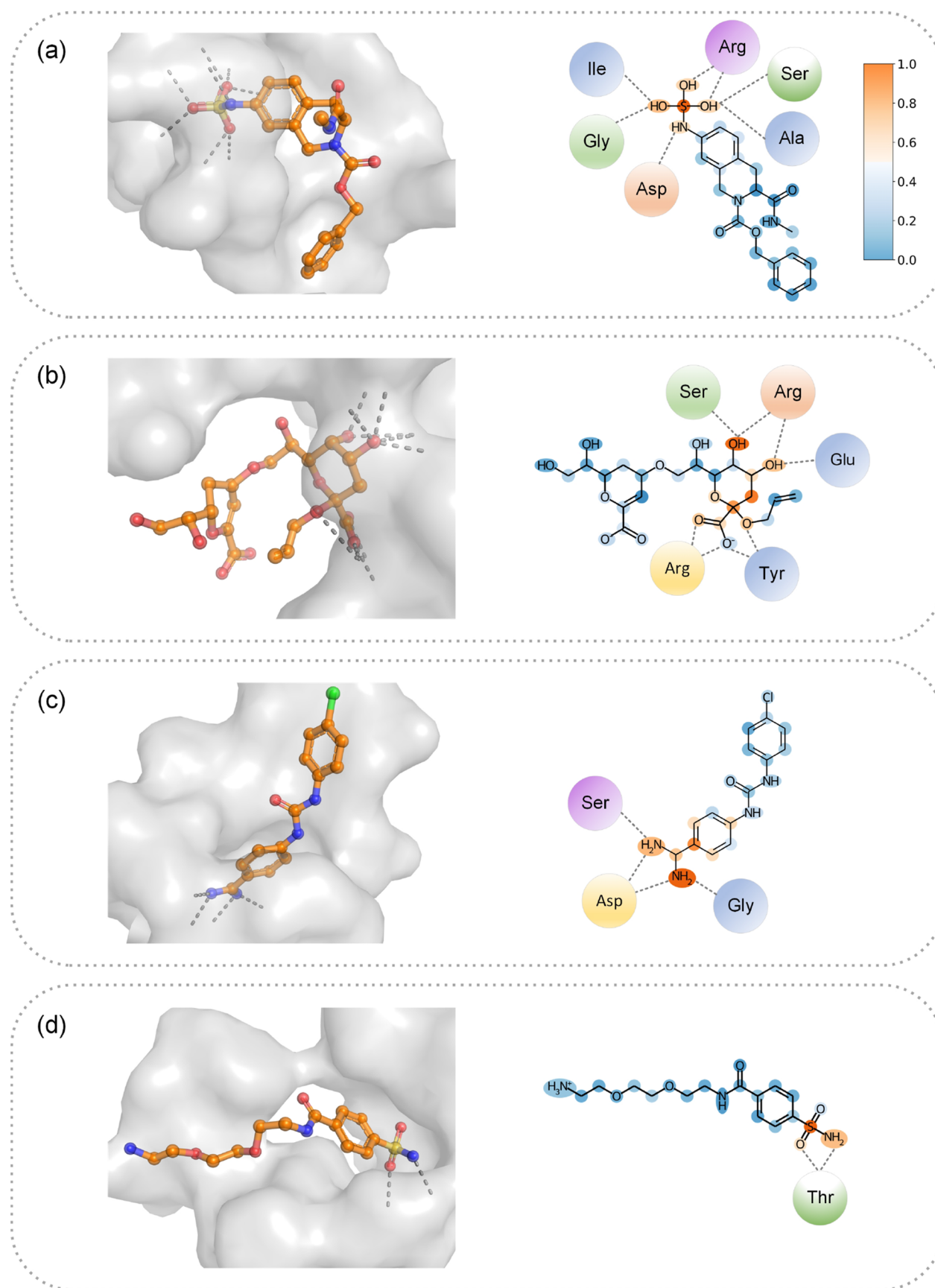


Figure 7. Visualizations of atom importance of ligands using Grad-AAM for complexes with PDB IDd (a) 2f6v, (b) 2r1x, (c) 1bjv, and (d) 1cnx, where the left and right columns are 3D and 2D binding modes with hydrogen bonds, respectively. The atom importance is shown with the blue/orange color in the 2D binding mode.

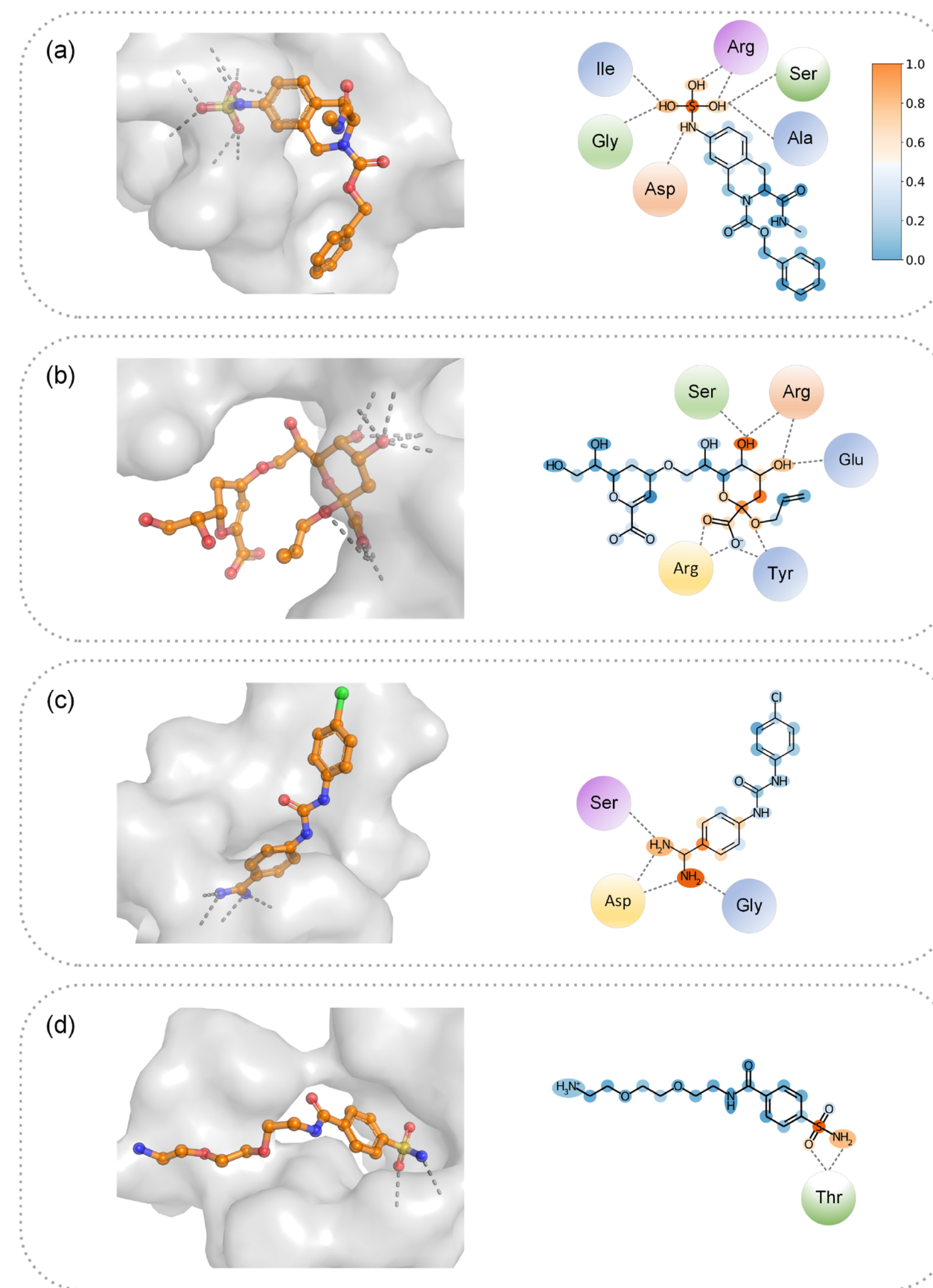


图7. 使用Grad-AAM对具有PDB ID (a) 2f6v, (b) 2r1x, (c) 1bjv和(d) 1cnx的复合物的配体原子重要性进行可视化, 其中左列和右列分别为显示氢键的三维和二维结合模式。在二维结合模式中, 原子重要性使用蓝色/橙色表示。

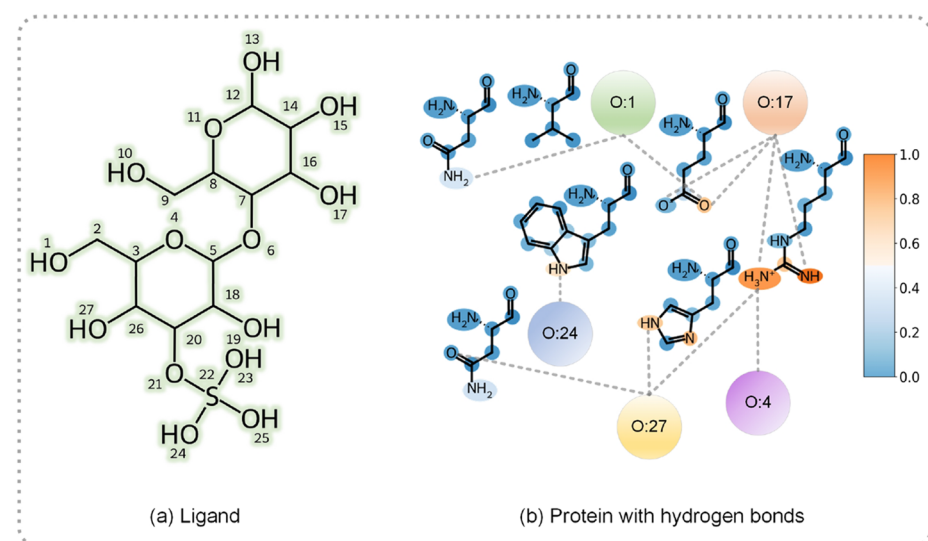


Figure 8. Visualizations of atom importance of the protein using Grad-AAM for the complex with PDB ID 4ym2. (a) The ligand with atom labels. (b) The protein with hydrogen bonds, where O:4 indicates the ligand atom O with atom label 4.

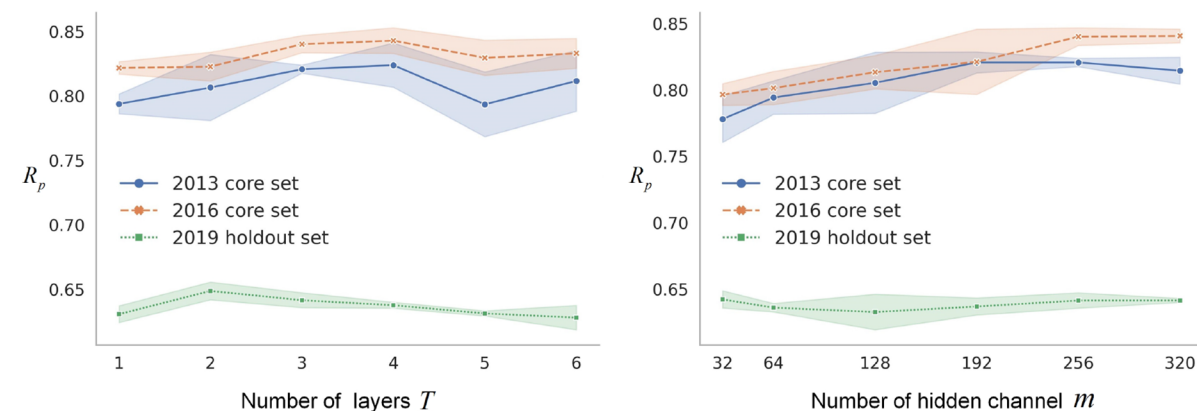


Figure 9. Analysis of parameters T and m on the three external test sets.

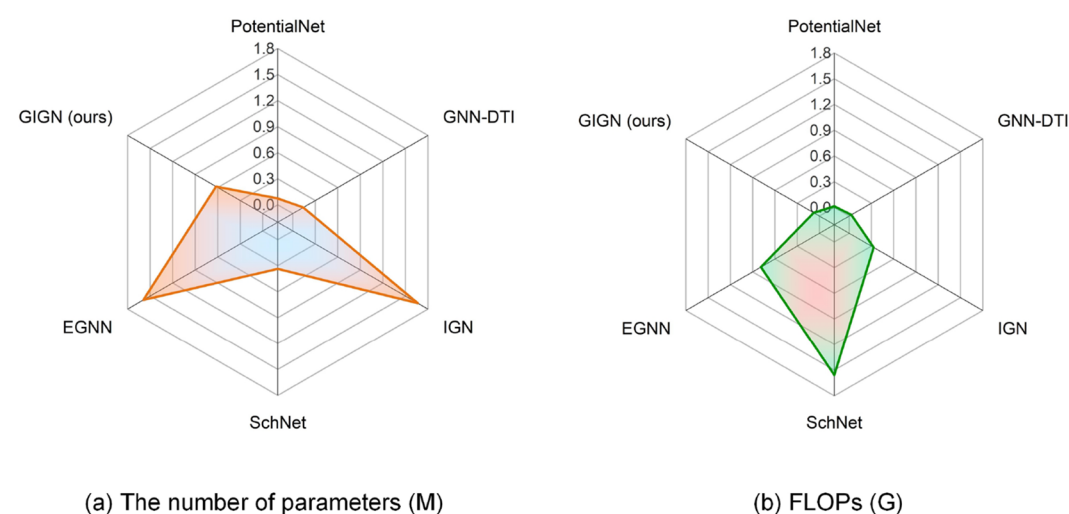


Figure 10. Model complexity in terms of (a) the number of parameters (the lower the better) and (b) FLOPs (the lower the better).

PLA from 3D structures and physical interactions. The core of GIGN is a heterogeneous interaction layer that unifies covalent and noncovalent interactions into the message passing phase while retaining the invariance property with respect to input

translations and rotations, thus avoiding the computational burden of data augmentation. Besides the better generalization capability than other baselines, the visualization results show that GIGN can capture essential features relevant to binding. We

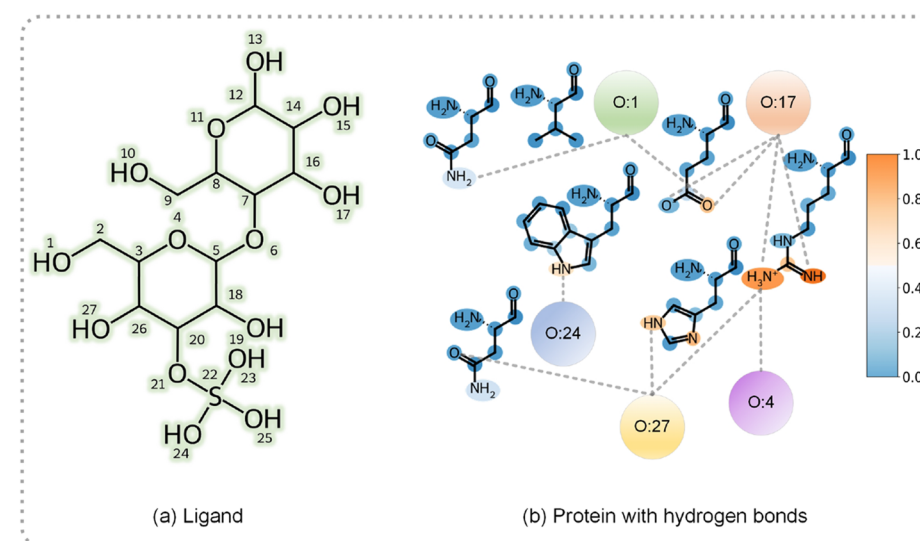


图 8. 使用Grad-AAM对PDB ID 4ym2的复合物进行的蛋白质原子重要性的可视化。(a) 带有原子标签的配体。(b) 带有氢键的蛋白质，其中O:4表示原子标签为4的配体原子O。

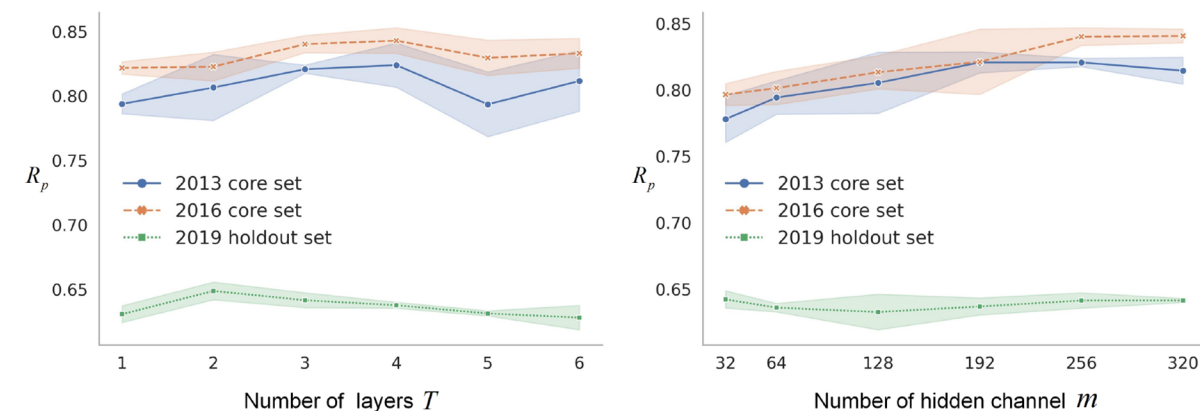


图 9. T 和 m 在三个外部测试集上的参数分析。

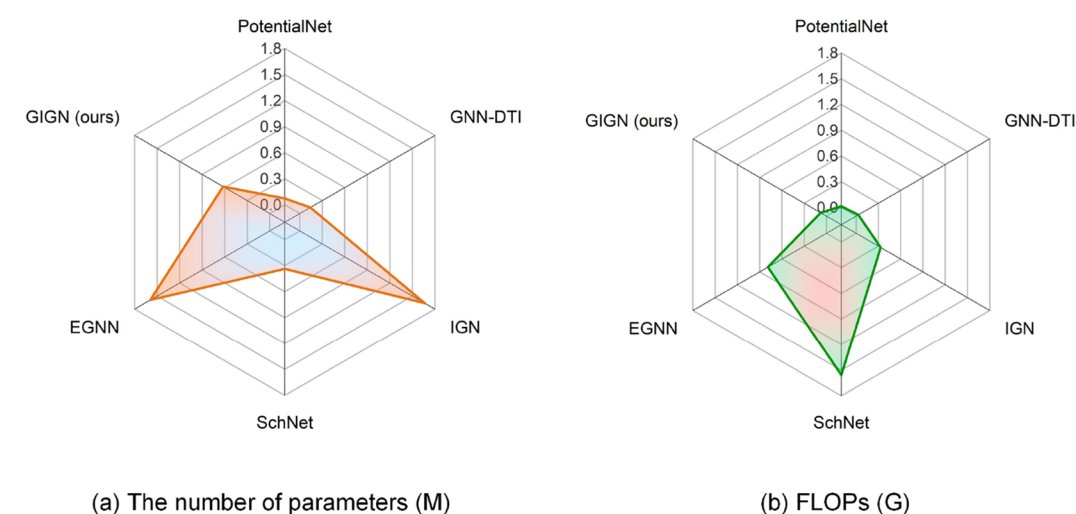


图 10. 模型复杂性方面的分析，包括(a) 参数数量（越低越好）和(b) FLOPs（越低越好）。

来自三维结构和物理交互的PLA。GIGN的核心是一个异质交互层，它将共价和非共价交互统一到信息传递阶段，同时保持对输入的不变性特性。

平移和旋转，从而避免数据增强的计算负担。除了较其他基准模型具有更好的泛化能力，可视化结果显示GIGN能够捕获与结合相关的基本特征。我们

hope that the proposed GIGN can inspire the design of other types of biological 3D interactions.

■ ASSOCIATED CONTENT

Data Availability Statement

Demo, instructions, code, and data for GIGN and other baselines are available at <https://github.com/guaguabujianle/GIGN>.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcllett.2c03906>.

Details of atom features, additional visualization results, and quantitative analysis of the number of covalent and noncovalent interactions (PDF)

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Notes

The authors declare no competing financial interest.

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希望所提出的GIGN能够激发其他类型生物三维相互作用的设计。

■ 关联内容

数据可用性声明

对GIGN和其他基准的演示、说明、代码和数据可在<https://github.com/guaguabujianle/GIGN>获取。

支持信息

支持信息可免费获得，网址为 <https://pubs.acs.org/doi/10.1021/acs.jpcllett.2c03906>。

原子特征的详细信息、额外的可视化结果，以及共价和非共价交互数量的定量分析 (PDF)

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备注

作者声明没有竞争性的经济利益。

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